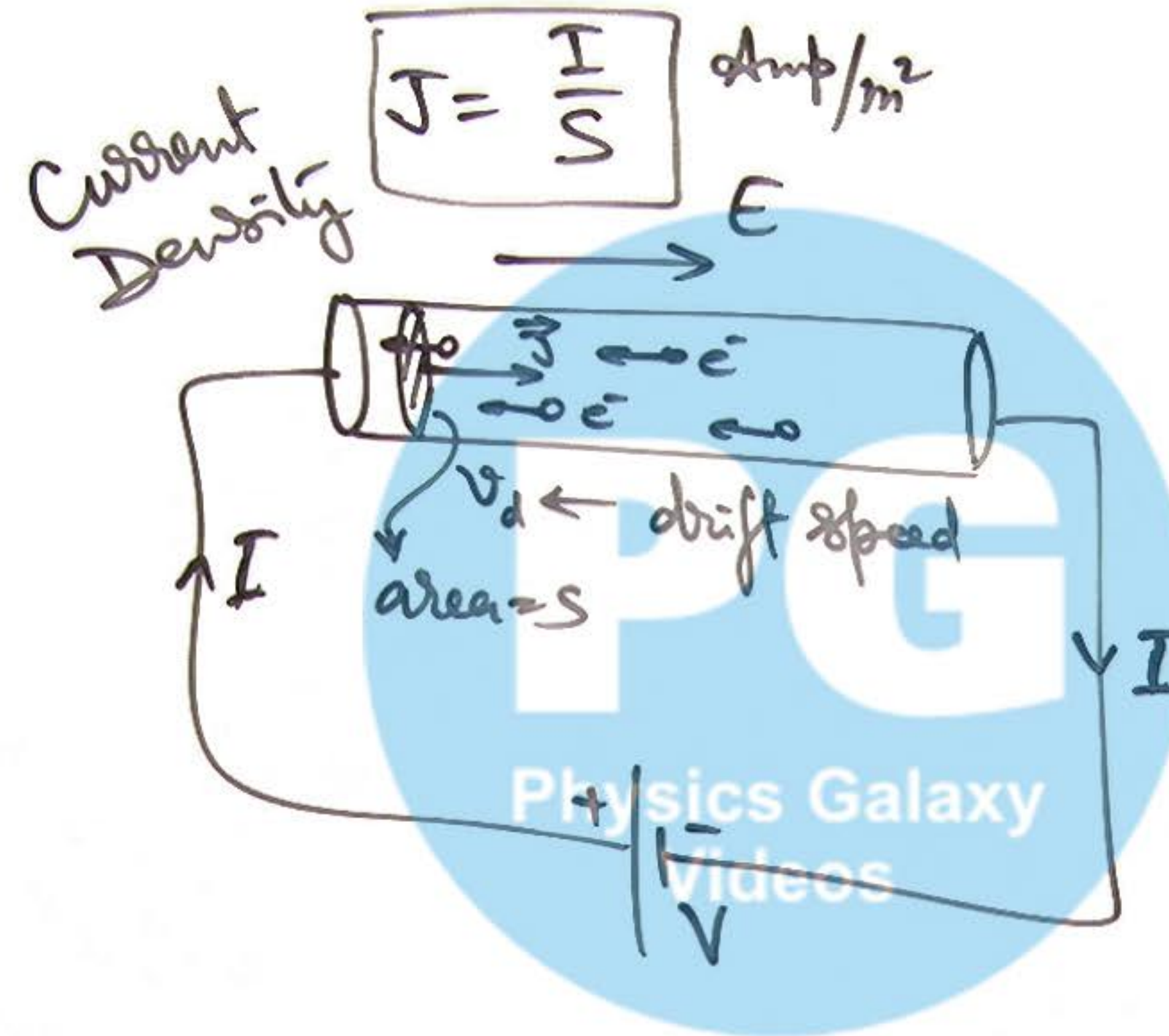


① Electric Current & Current Density:



$$v_d = \mu E \quad (\text{avg relaxation})$$

$$v_d = \frac{e E \tau}{m}$$

$$J = n e v_d \quad \rightarrow \text{free } e^- \text{ density in conductor}$$

$$J = \frac{n e^2 E \tau}{m}$$

② Ohm's Law:

if l & S are constant
 $R = R_0(1 + \alpha \Delta T)$

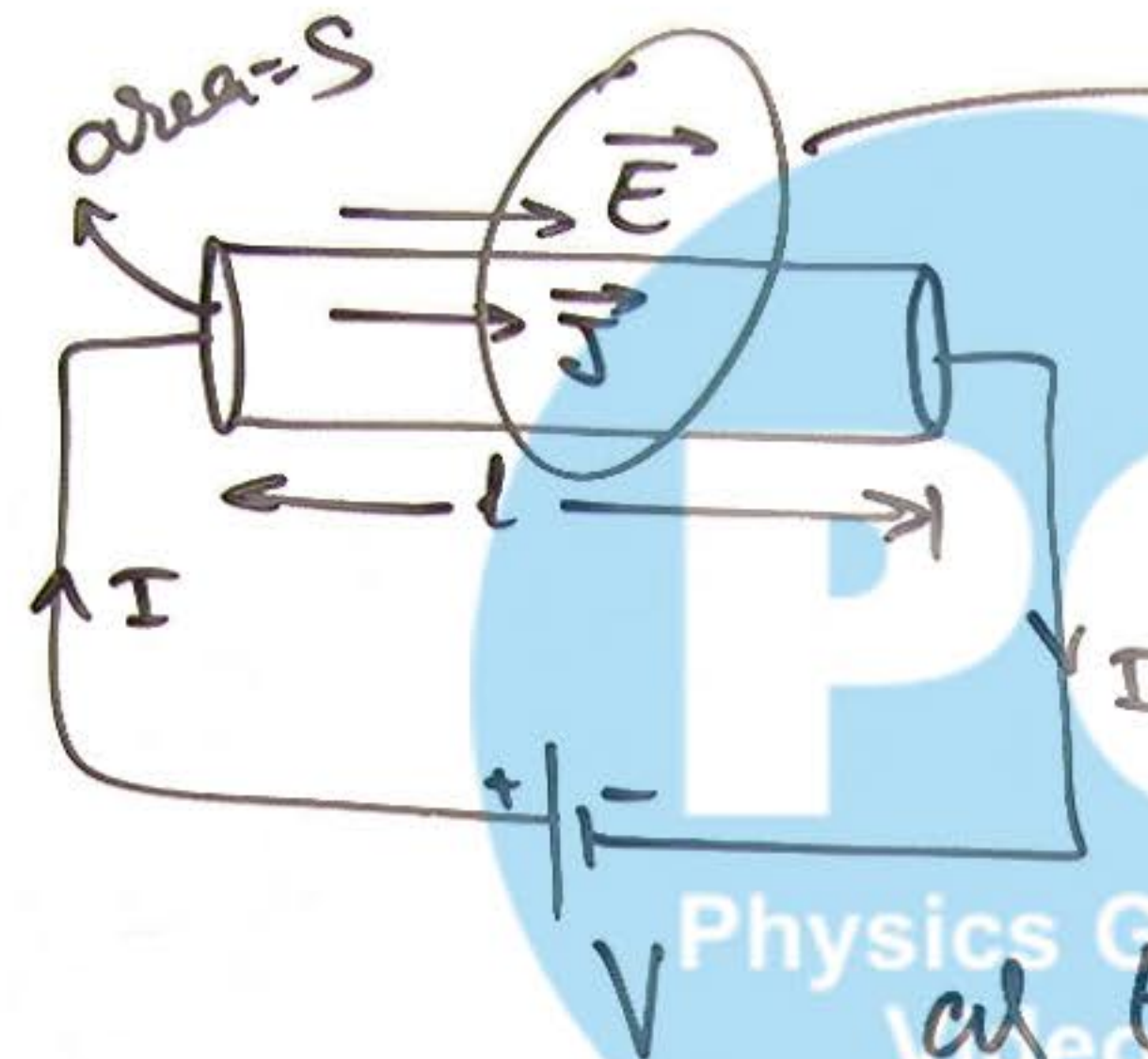
Resistance

$$R = \frac{\rho l}{S}$$

ρ depends on temp ✓

$$\rho = \rho_0(1 + \alpha \Delta T)$$

temp. Coeff of Resistance



$$\vec{J} = \sigma \vec{E}$$

Conductivity

Resistivity $\rho = \frac{1}{\sigma}$

$$\text{or } \vec{E} = \rho \vec{J} \checkmark$$

Differential form of Ohm's law.

$$\text{as } E = \frac{dV}{dl}$$

$$\frac{dV}{dl} = \rho \frac{I}{S}$$

$$\int dV = \frac{\rho I}{S} \int dl$$

$$V = \left(\frac{\rho l}{S} \right) I$$

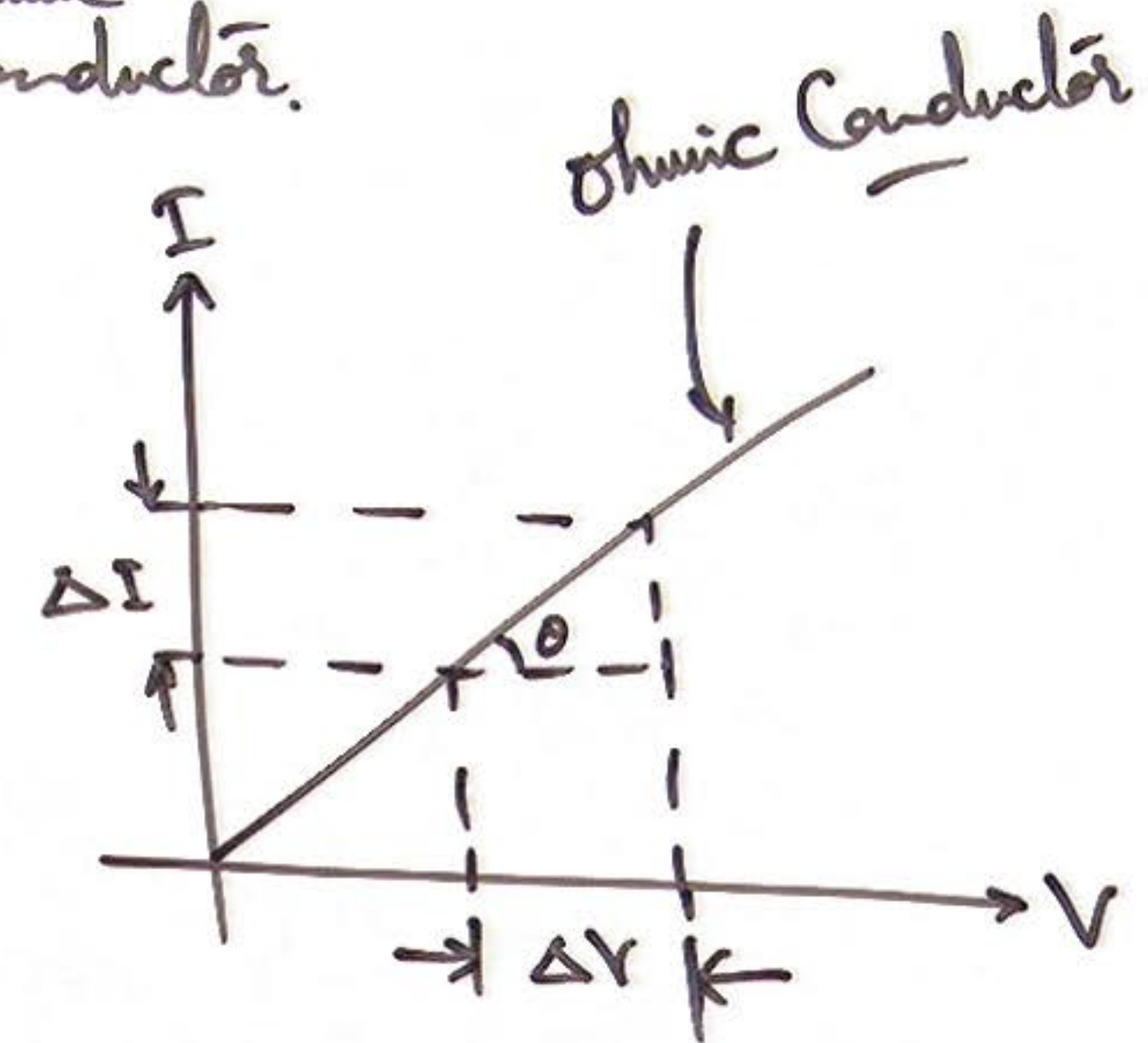
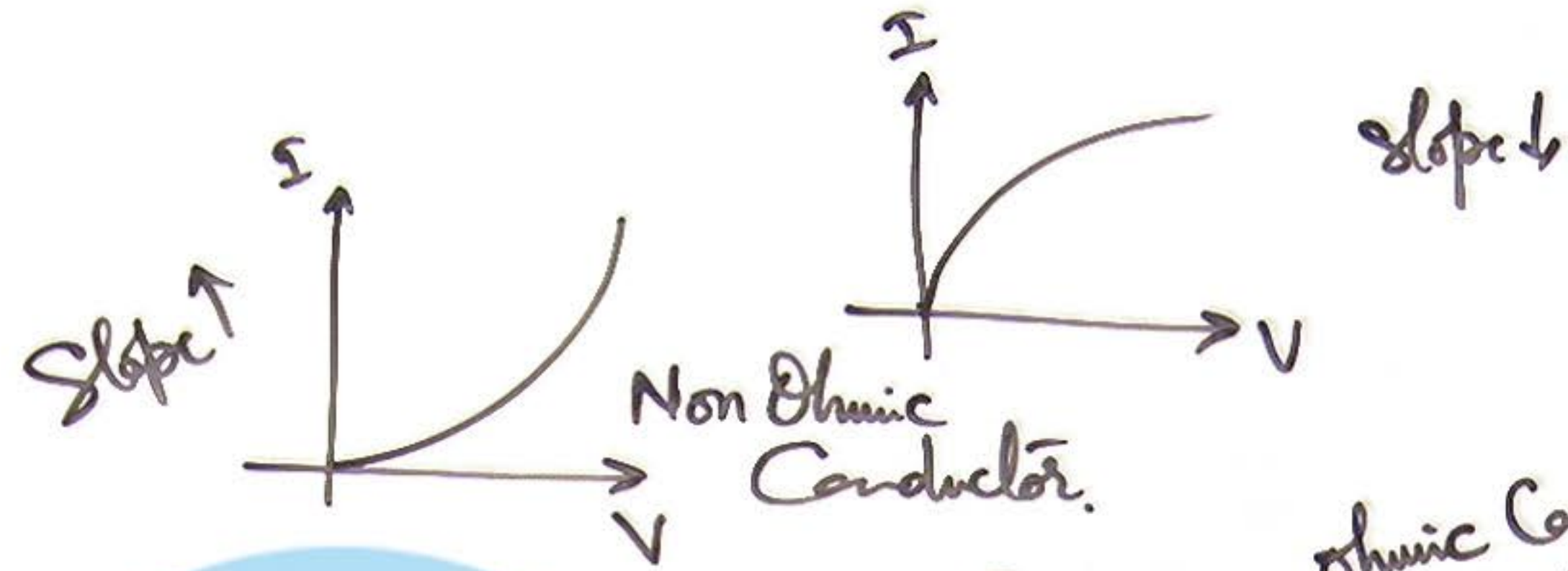
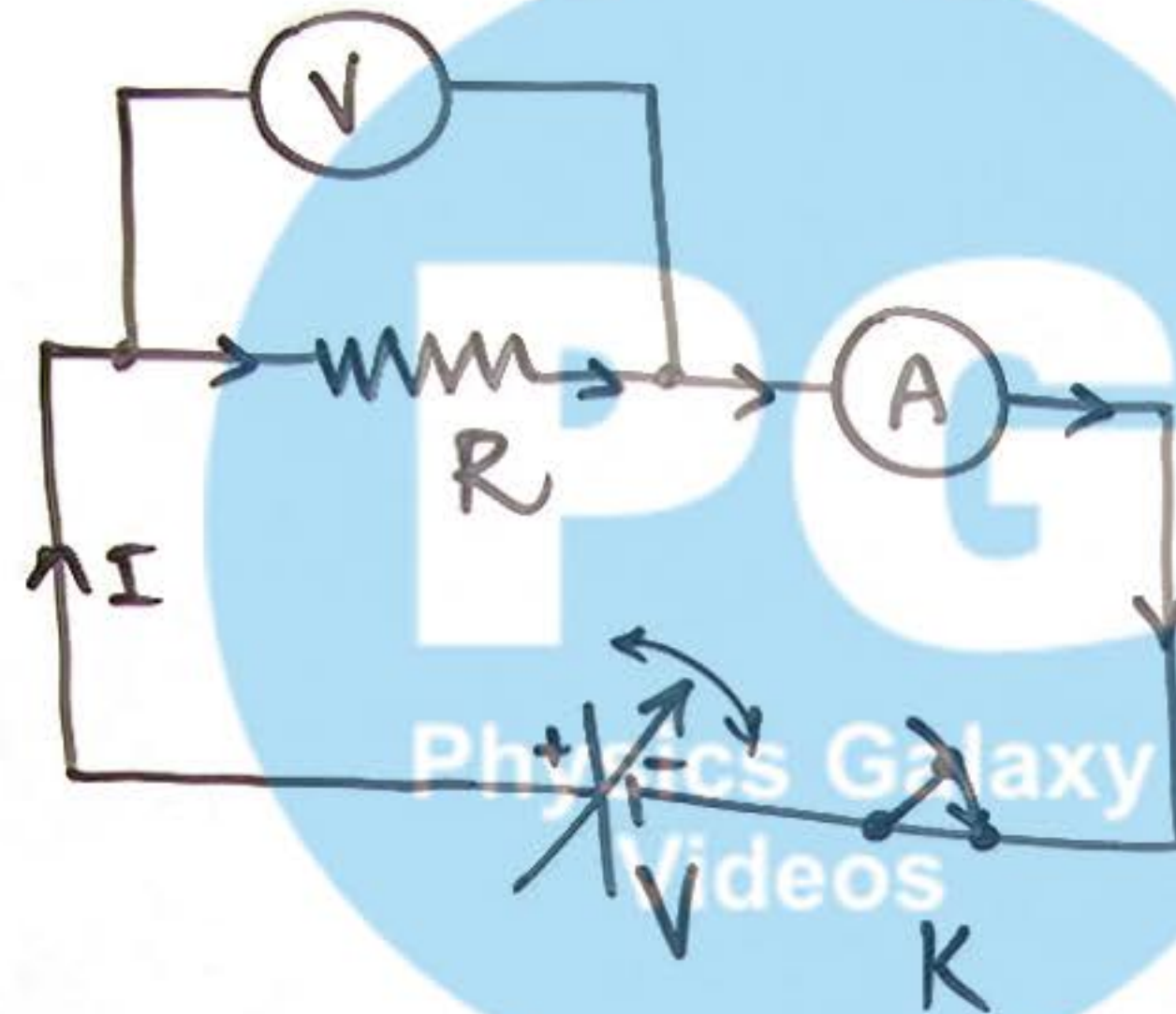
$$V = IR$$

③ Verification of Ohm's Law:

$$V = IR$$

$$\Delta V = \Delta I R$$

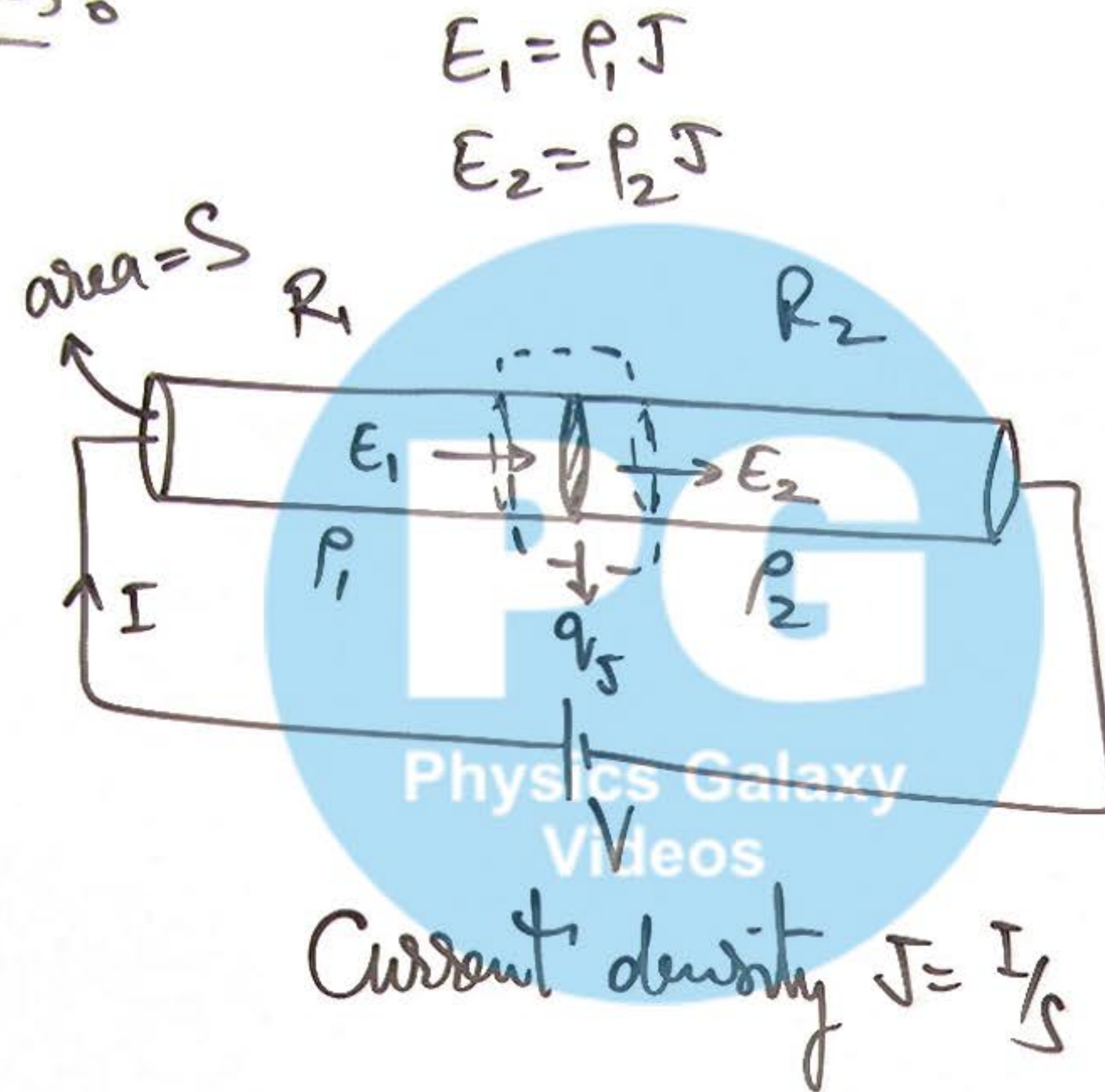
$$\frac{\Delta I}{\Delta V} = \frac{1}{R}$$



$$\text{Slope}(\tan \theta) = \frac{\Delta I}{\Delta V}$$

④ Charge accumulation at Junction of Two conductors:

$$I = \frac{V}{R_1 + R_2}$$



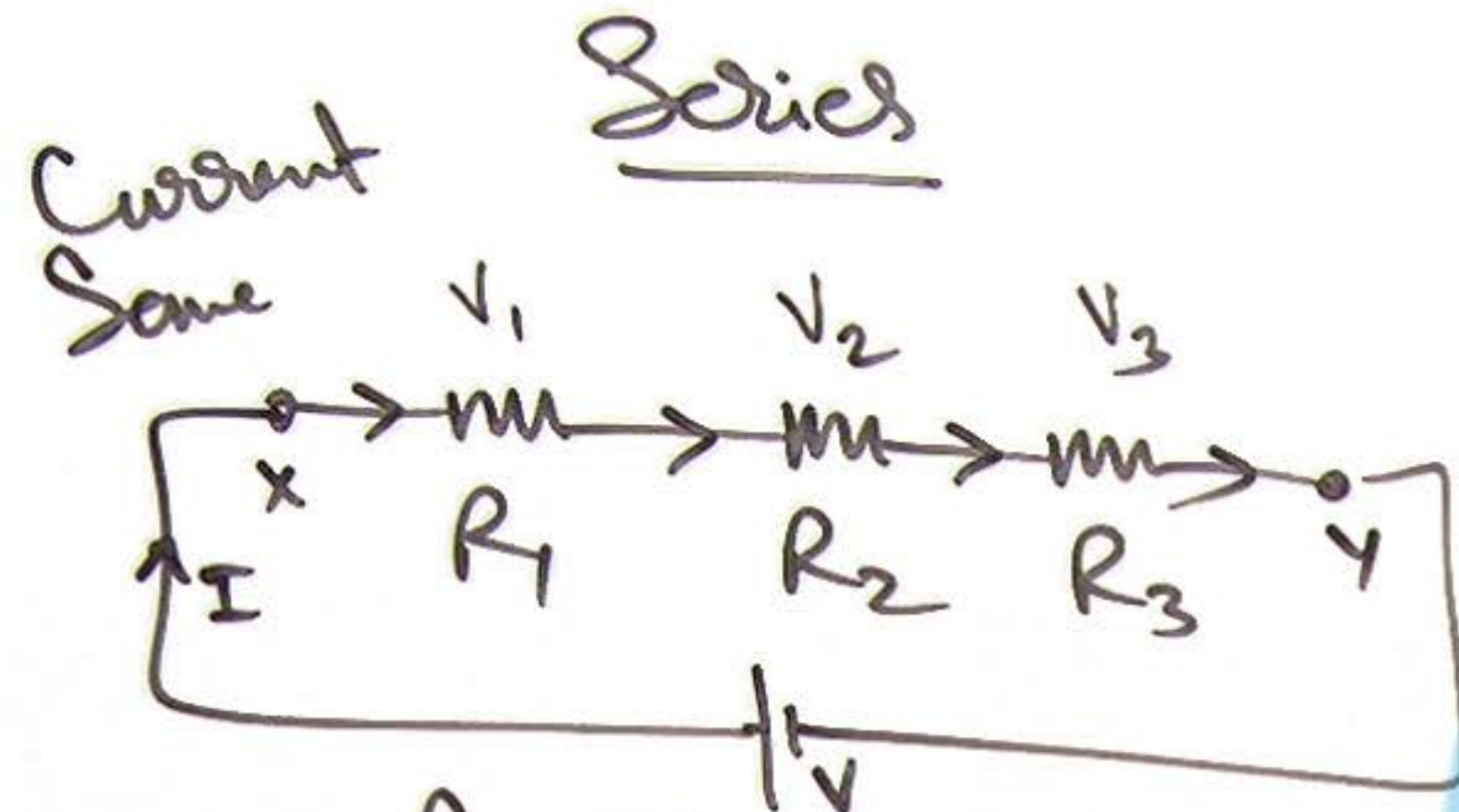
Using Gauss's law.

$$E_2 \cdot S - E_1 \cdot S = \frac{q_J}{\epsilon_0}$$

$$\rho_2 I - \rho_1 I = \frac{q_J}{\epsilon_0}$$

$$\Rightarrow \boxed{q_J = \epsilon_0 (\rho_2 - \rho_1) I}$$

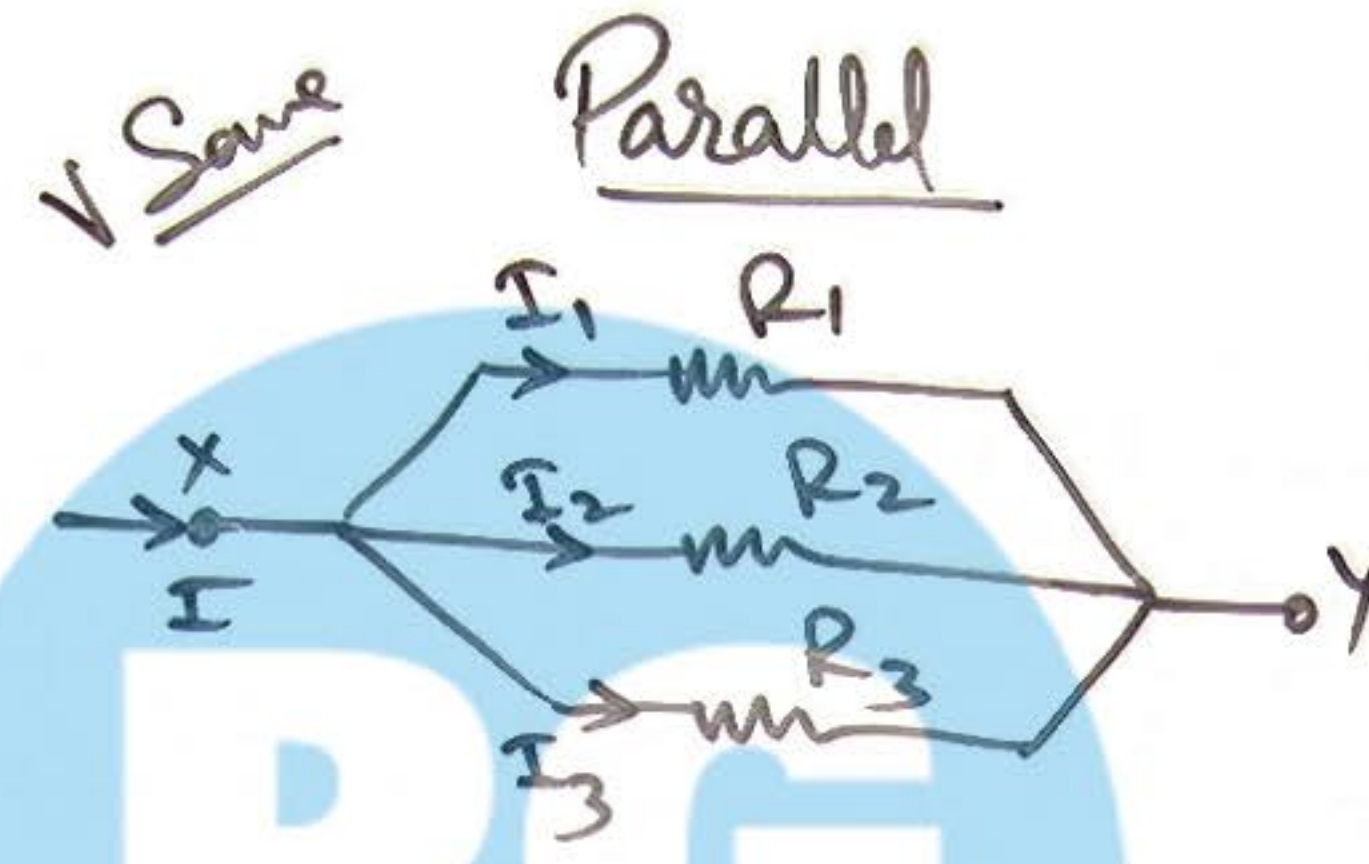
⑤ Combination of Resistances:



$$R_{xy} = R_1 + R_2 + R_3$$

if all Res are equal & there are N -resistance

$$R_{eq} = NR \rightarrow V_{each} = \frac{V}{N}$$

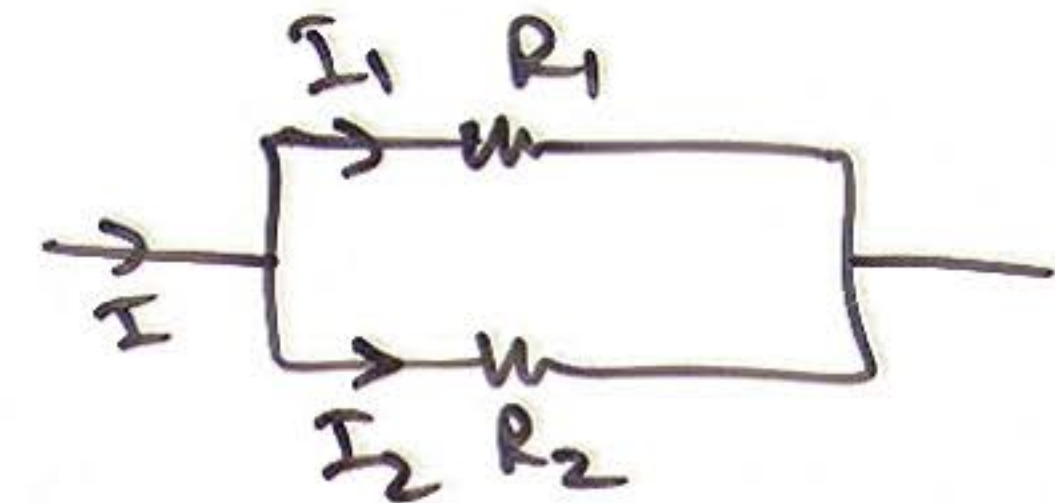


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

if all Res are same (R) & there are N -resistance

$$R_{eq} = R/N \rightarrow I_{each} = \frac{I}{N}$$

$$I_1 = \frac{IR_2}{R_1 + R_2} \quad \& \quad I_2 = \frac{IR_1}{R_1 + R_2}$$



$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

⑥ Resistance with parameter variation:

The diagram illustrates two scenarios of resistance calculation when a material property (resistivity) varies along the length of a conductor.

Left Diagram (Cylindrical Wire): A cylinder of length l and cross-sectional area S is shown. The resistivity ρ varies along the x -axis, denoted as $\rho = f(x)$. A small element of length dx is highlighted. The total resistance is given by the formula:

$$R = \frac{\rho l}{S}$$

Right Diagram (Rectangular Block): A rectangular block of length l and cross-sectional area $w \cdot y$ is shown. The resistivity ρ varies along the y -axis, denoted as $\rho = f(y)$. A small element of width dy is highlighted. The differential resistance dR for this element is:

$$dR = \frac{\rho(y) \cdot l}{w \cdot dy}$$

The total resistance R_T is found by integrating the differential resistance:

$$\frac{1}{R_T} = \int \frac{1}{dR} = \int_0^h \frac{w \cdot dy}{l \cdot \rho(y)}$$

Integration Formulas:

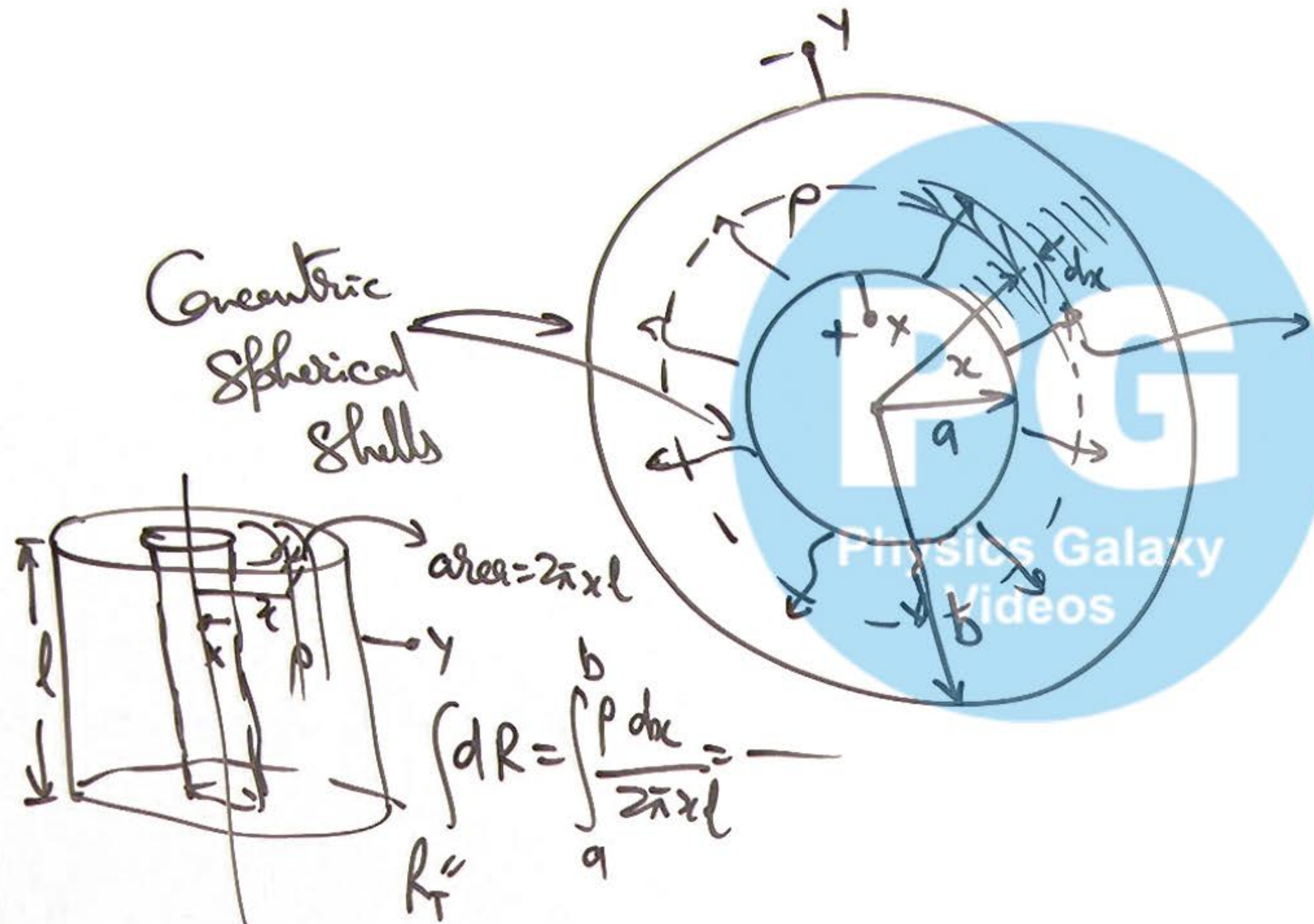
For the cylindrical wire, the total resistance is:

$$R_{\text{Total}} = \int dR = \int_0^l \frac{\rho(x) dx}{S}$$

For the rectangular block, the total resistance is:

$$R_{\text{Total}} = \int dR = \int_0^h \frac{\rho(y) \cdot l}{w \cdot dy}$$

⑦ Resistance for Radial Conduction:

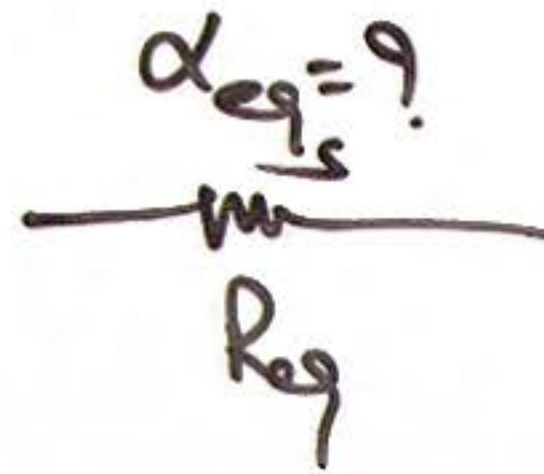
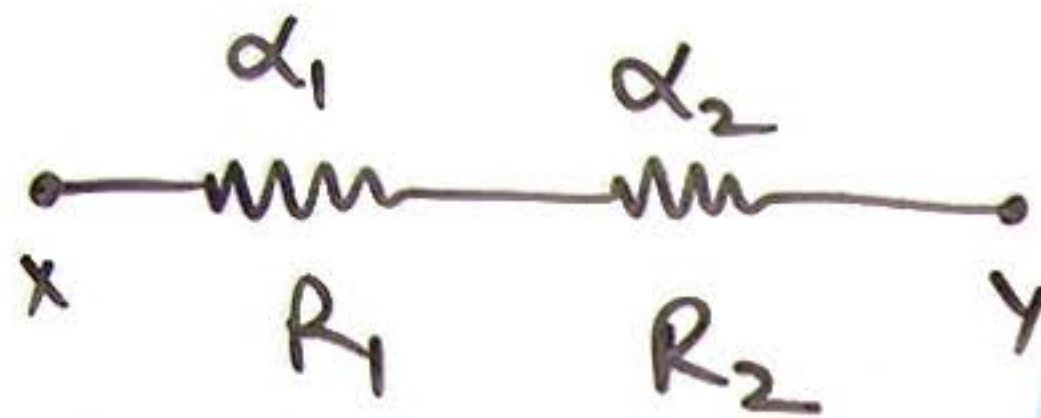


Resistance $dR = \frac{\rho dx}{4\pi x^2}$

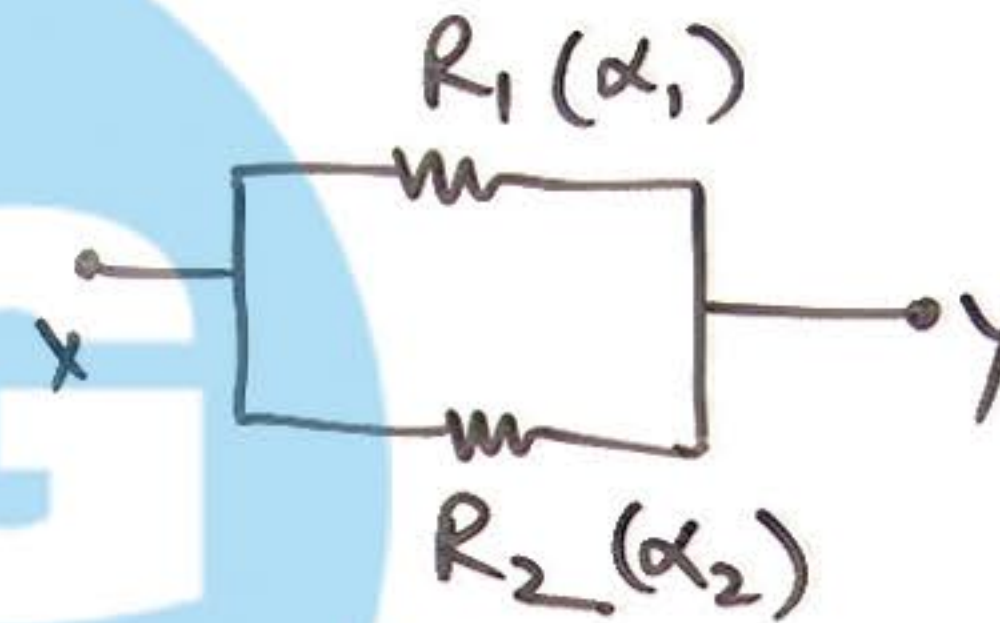
$$R_{Txy} = \int dR = \int_a^b \frac{\rho}{4\pi} \cdot \frac{dx}{x^2}$$

$$R_{xy} = \frac{\rho}{4\pi} \left[-\frac{1}{x} \right]_a^b = \frac{\rho}{4\pi} \left[\frac{1}{a} - \frac{1}{b} \right]$$

⑧ Equivalent Temp. Coefficient
of Resistance for combinations:

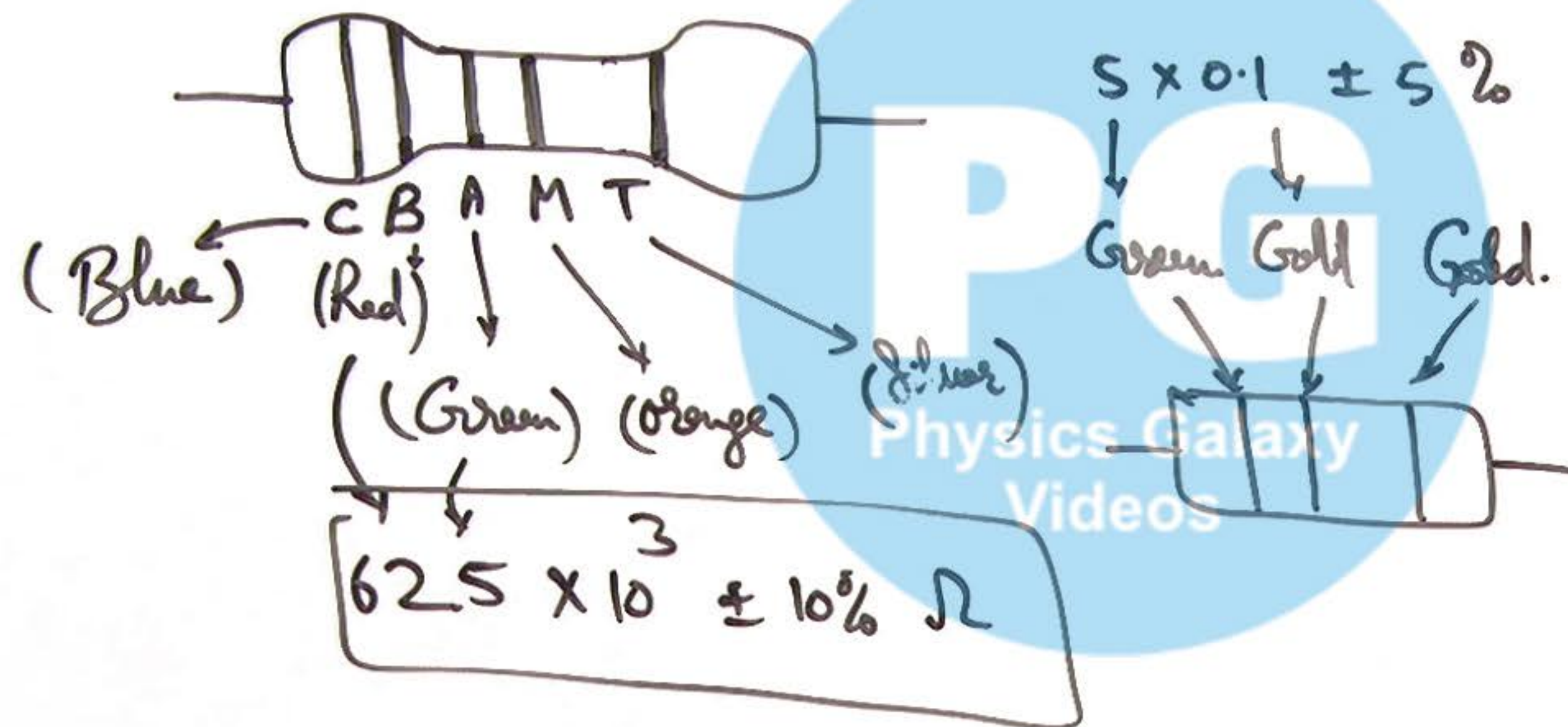


$$\alpha_{eq} = \alpha_s = \frac{\alpha_1 + \alpha_2}{2}$$



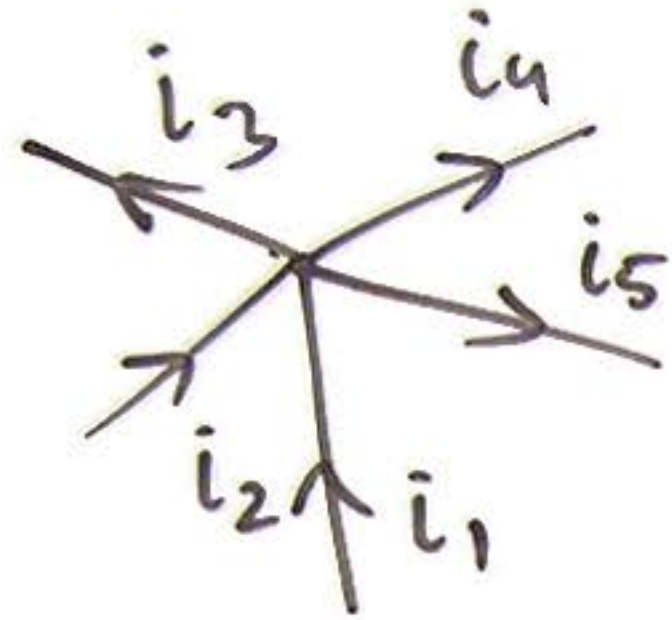
⑨ Colour Coding of Resistances:

Q1: Find Colour Code of $0.5\Omega \pm 5\%$



Colour	Colour Code		
Black	— 0		
Brown	— 1		
Red	— 2		
Orange	— 3		
Yellow	— 4		
Green	— 5		
Blue	— 6		
Violet	— 7		
Grey	— 8		
White	— 9		
Gold		M	(%)
Silver		T	
None			

⑩ Kirchoff's Current Law: (KCL)

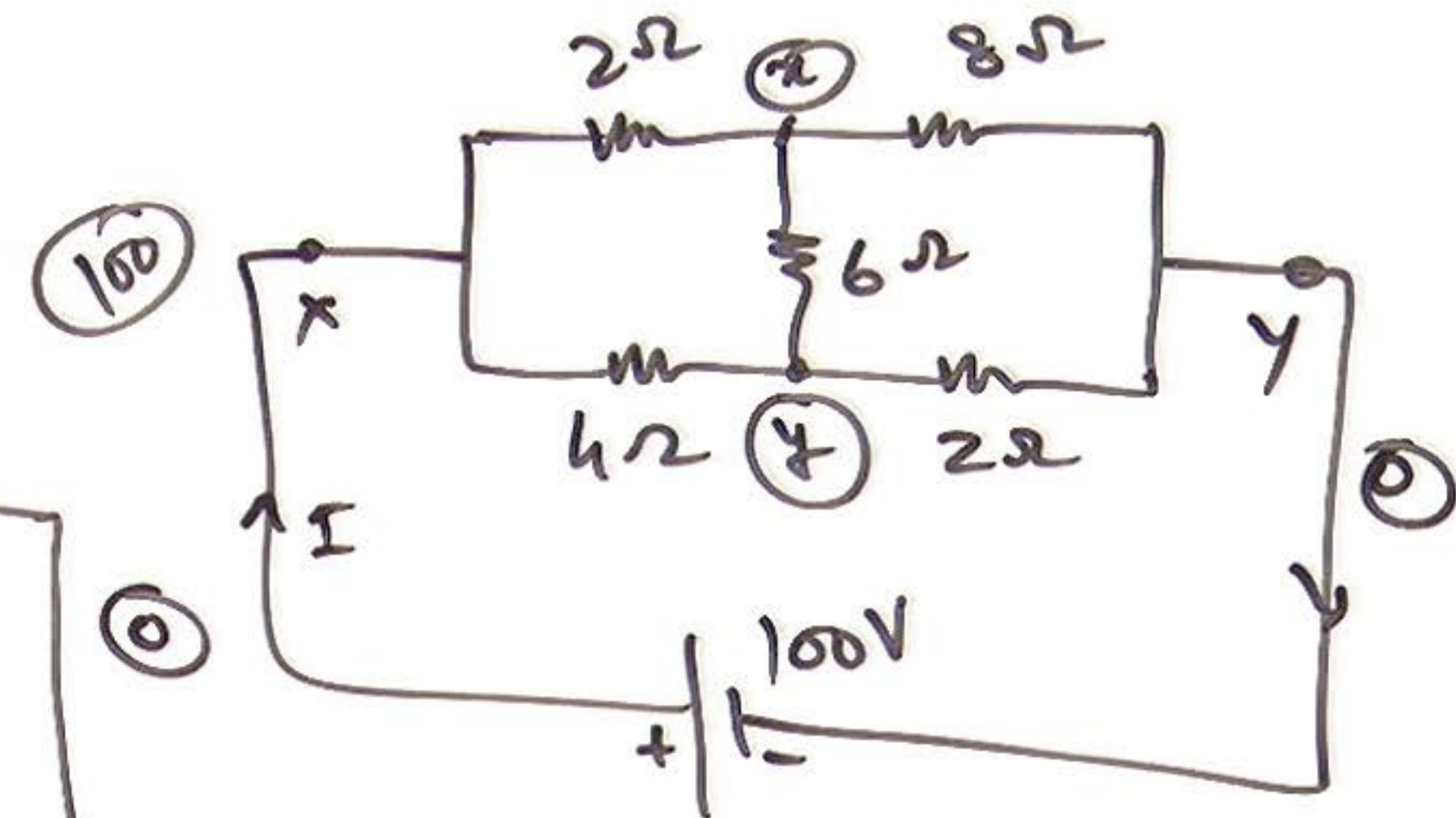


KCL

$$\text{eqn} [i_3 + i_4 + i_5 - i_1 - i_2 = 0]$$

$$R_{eq} = \frac{V}{I} = \frac{100}{\frac{x}{8} + \frac{y}{2}} \checkmark$$

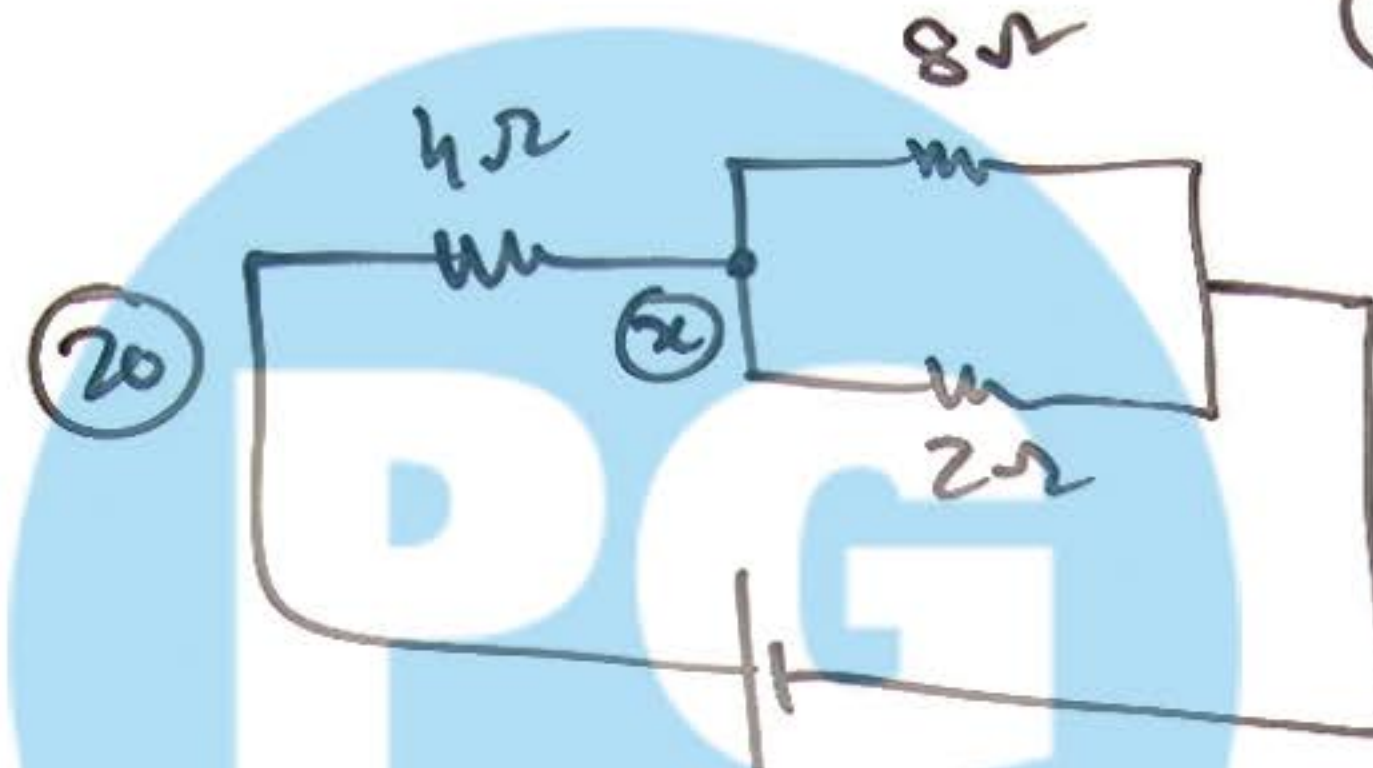
⑪ Wheatstone Bridge:



$$\text{KCL } \textcircled{x} \rightarrow \frac{x}{8} + \frac{x-y}{6} + \frac{x-100}{2} = 0$$

$$\text{KCL } \textcircled{y} \rightarrow \frac{y}{2} + \frac{y-x}{6} + \frac{y-100}{4} = 0$$

$$\Rightarrow \begin{matrix} x = \text{---} \\ y = \text{---} \end{matrix} \checkmark$$

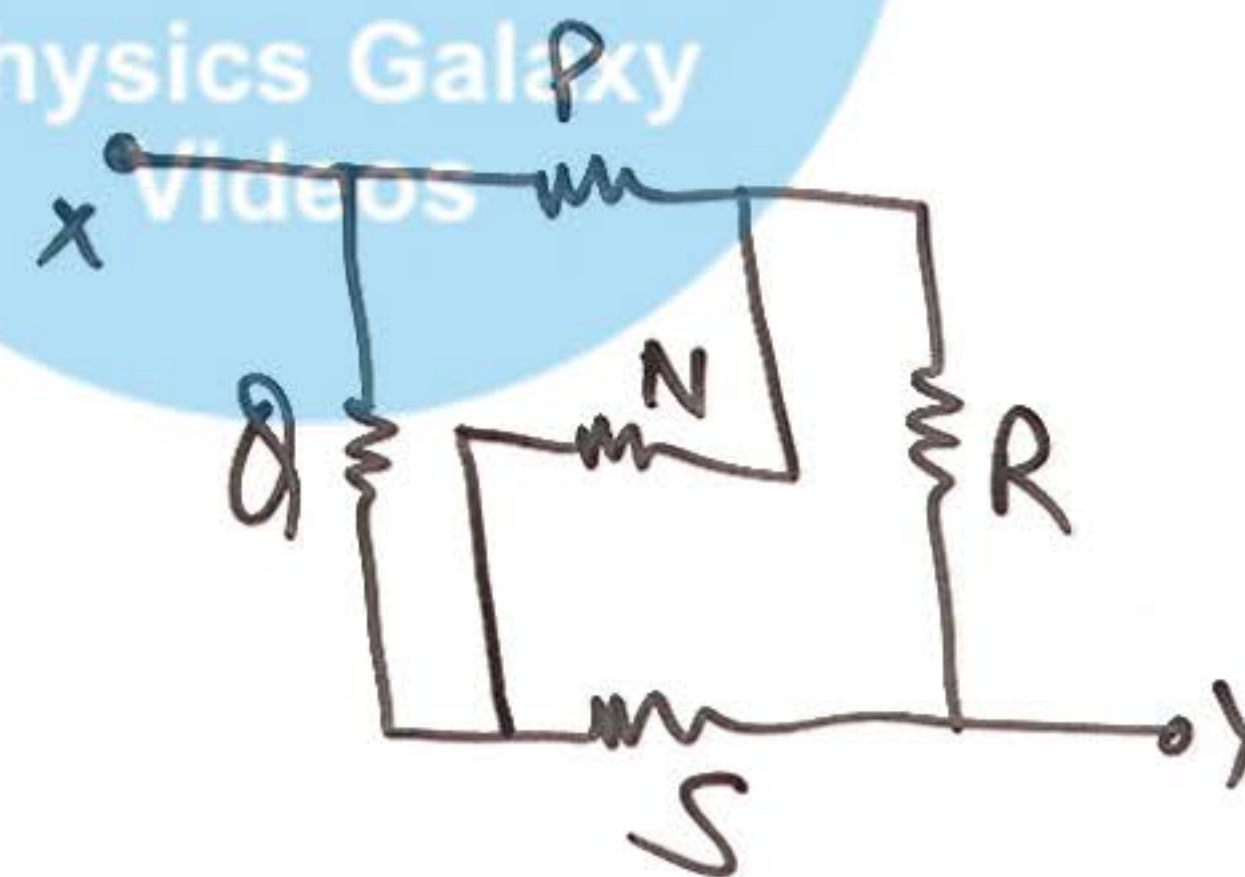
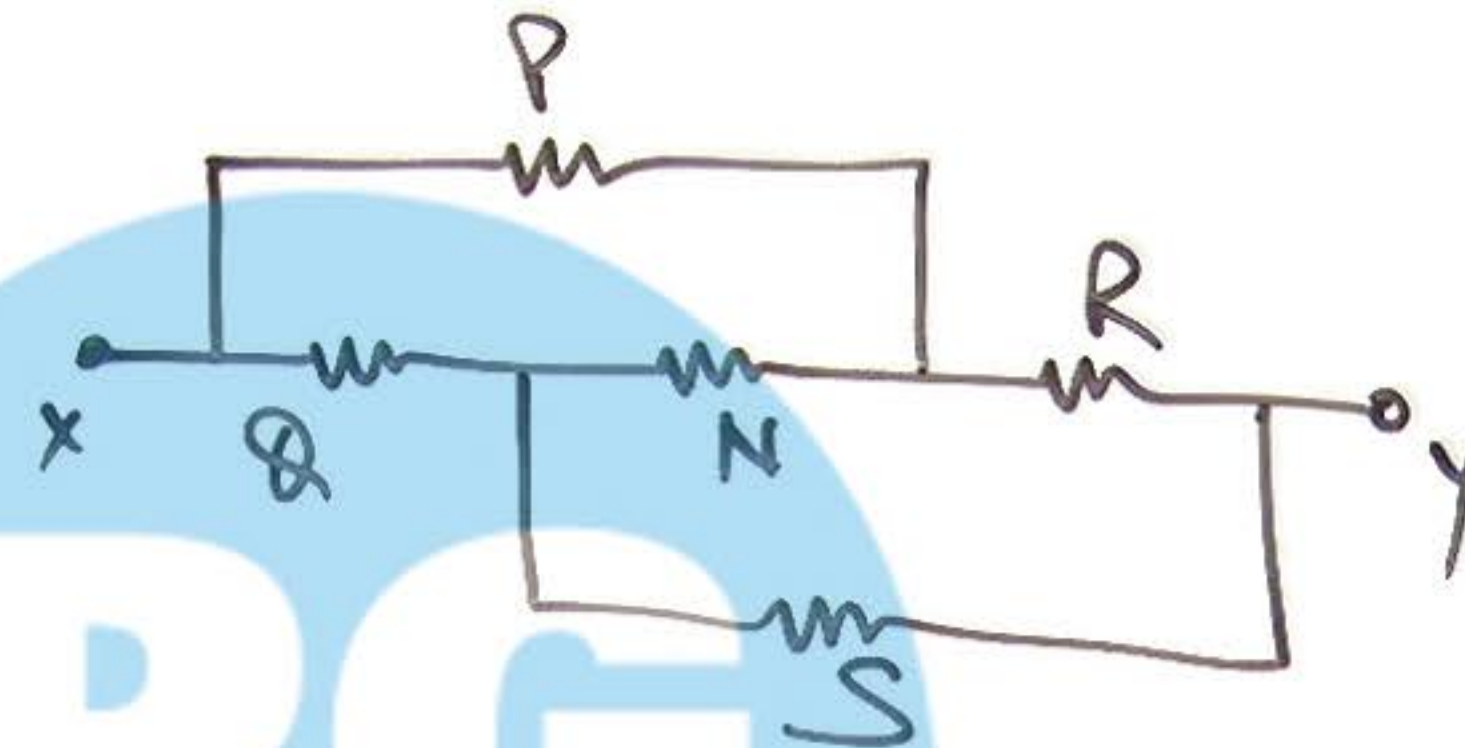
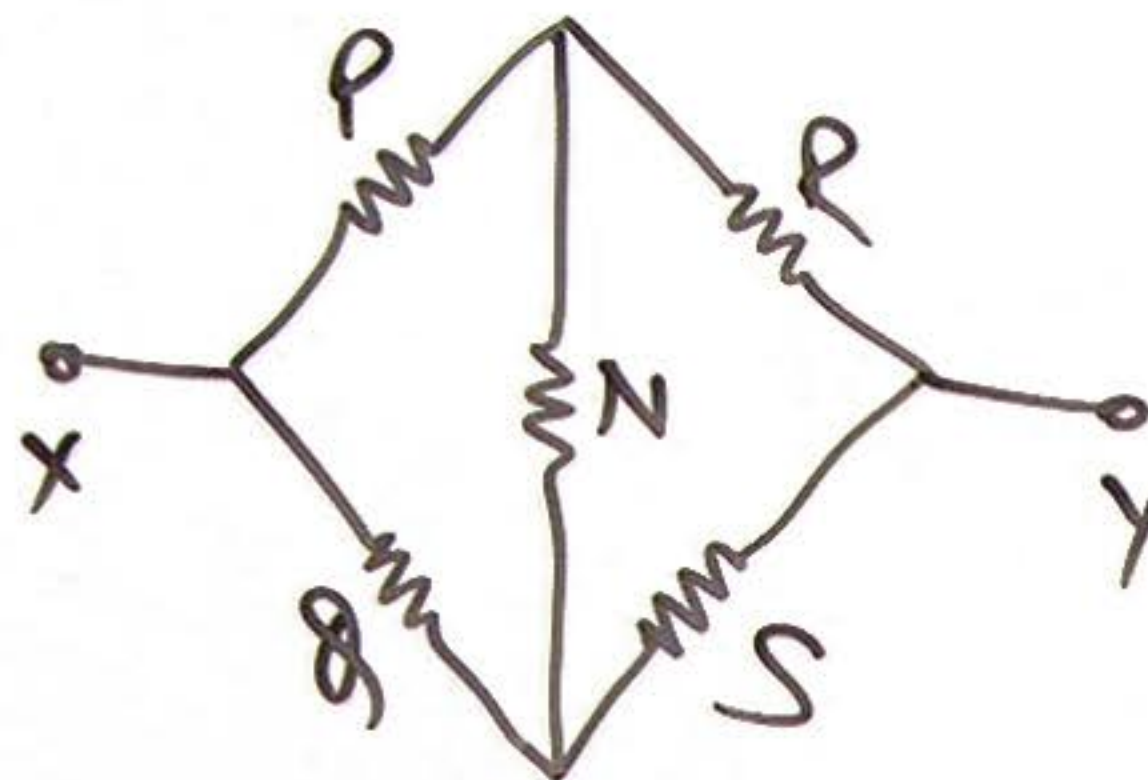
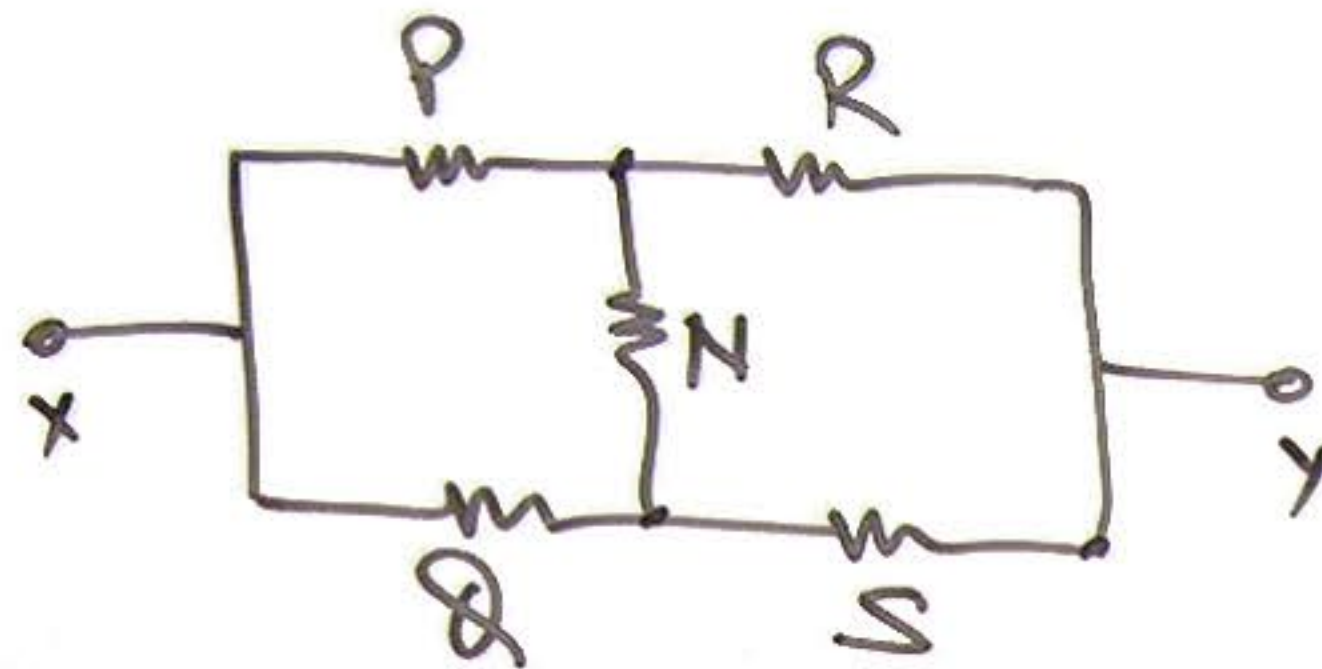


KCL eqn for $\textcircled{x}$

$$\frac{x}{8} + \frac{x}{2} + \frac{x-20}{4} = 0$$

$$x = \text{---}$$

⑫ Balancing of Wheatstone Bridge :



Balancing
Condition of Ws Bridge

$$\frac{P}{Q} = \frac{R}{S}$$

13 Analysis of Circuit Branches:

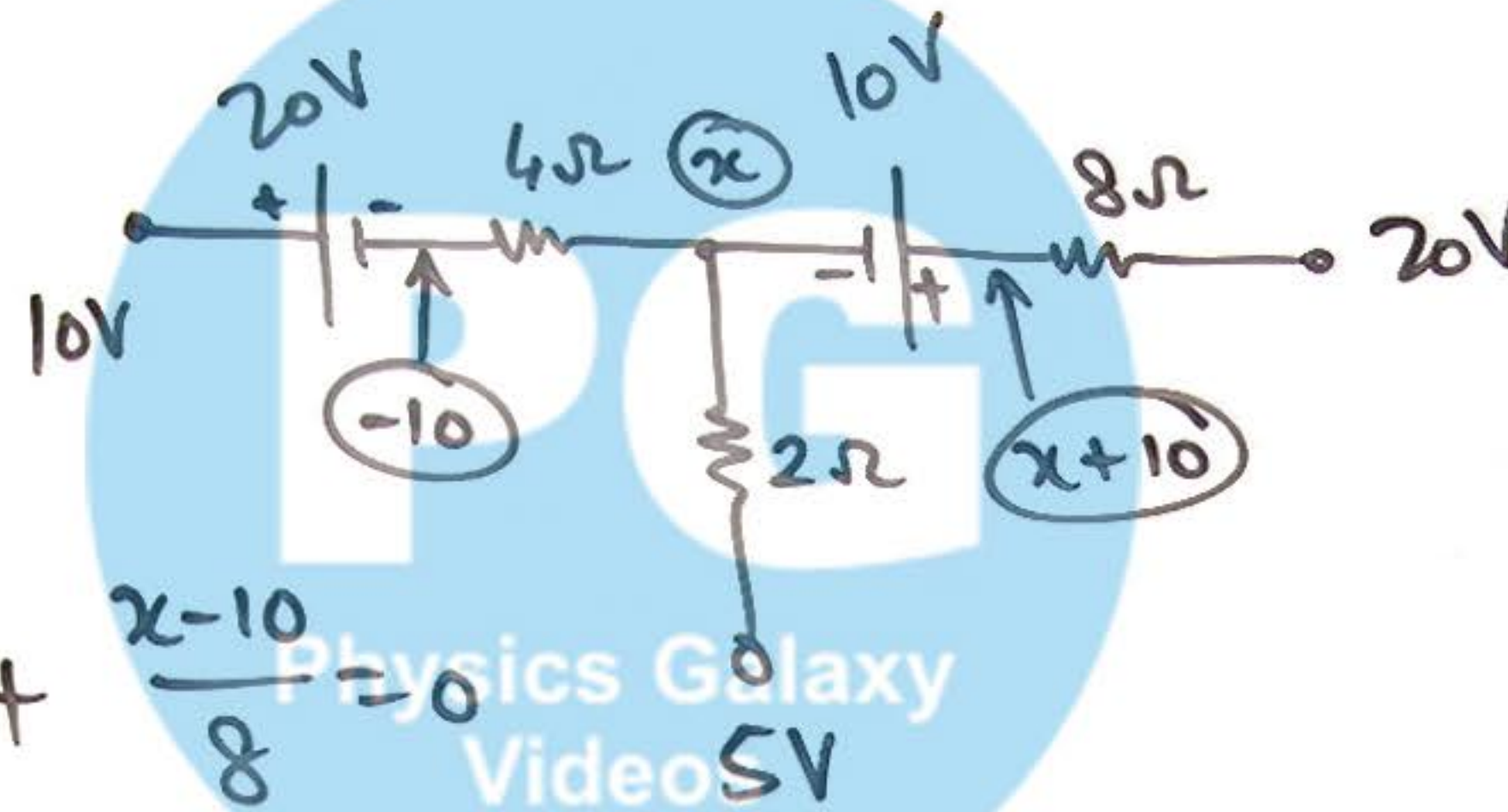
KCL eqⁿ for (x)

$$\frac{x-5}{2} + \frac{x+10}{4} + \frac{x-10}{8} = 0$$

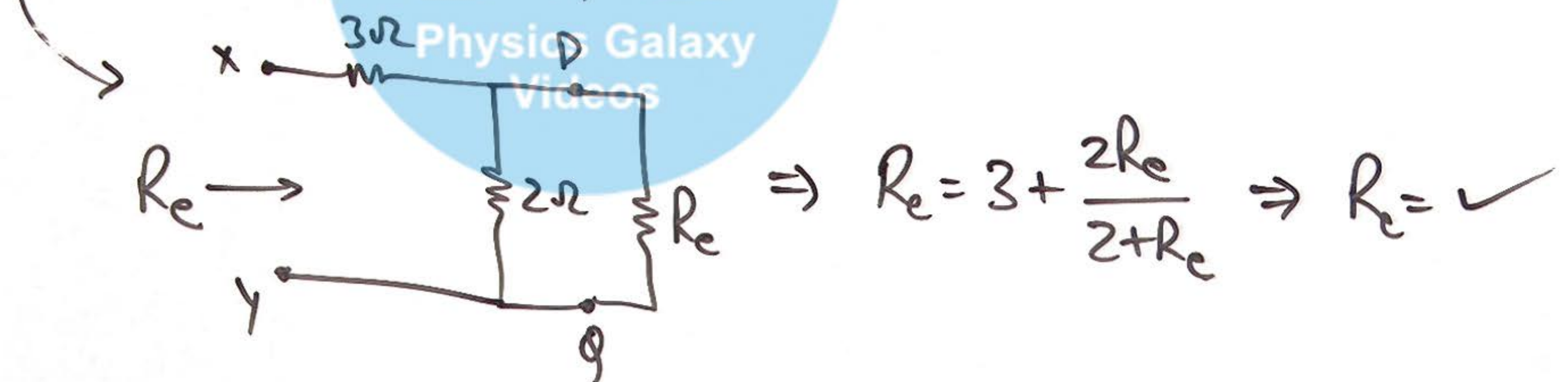
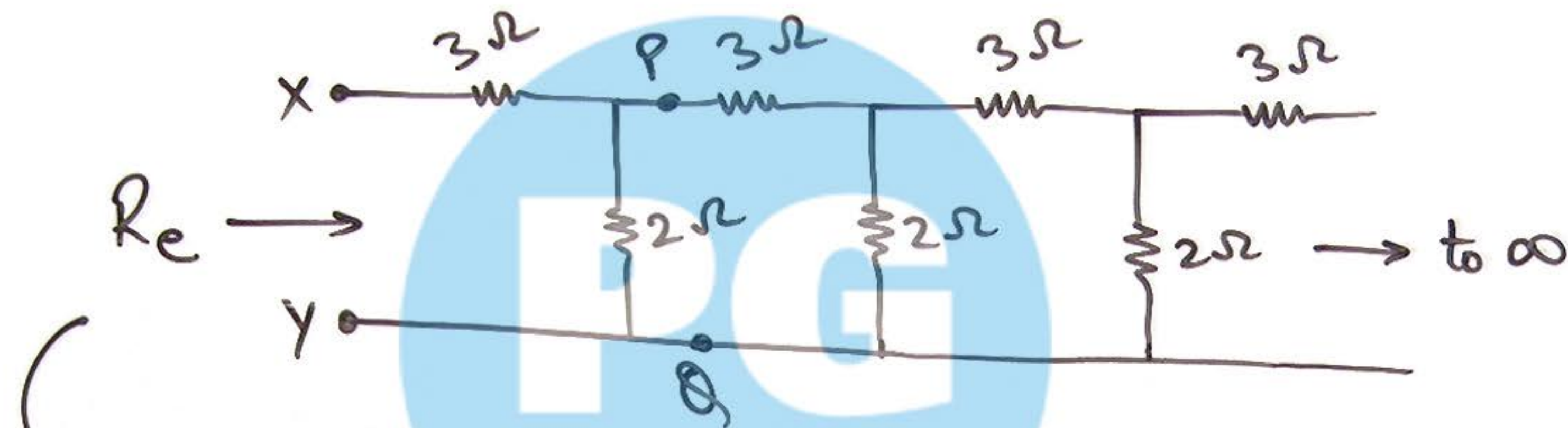
$$4x - 20 + 2x + 20 + x - 10 = 0$$

$$7x = 10$$

$$x = 10/7 \text{ volt}$$

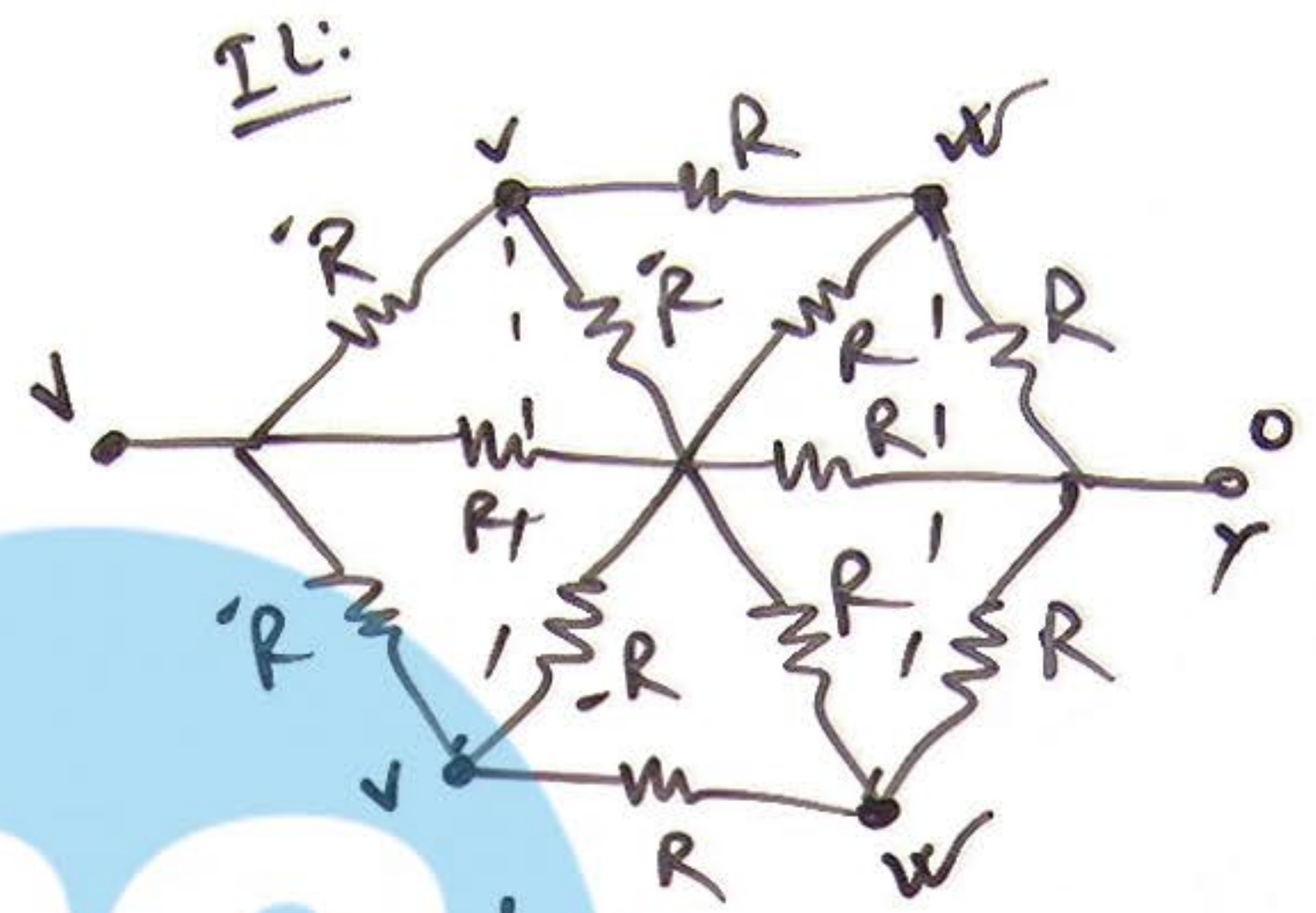
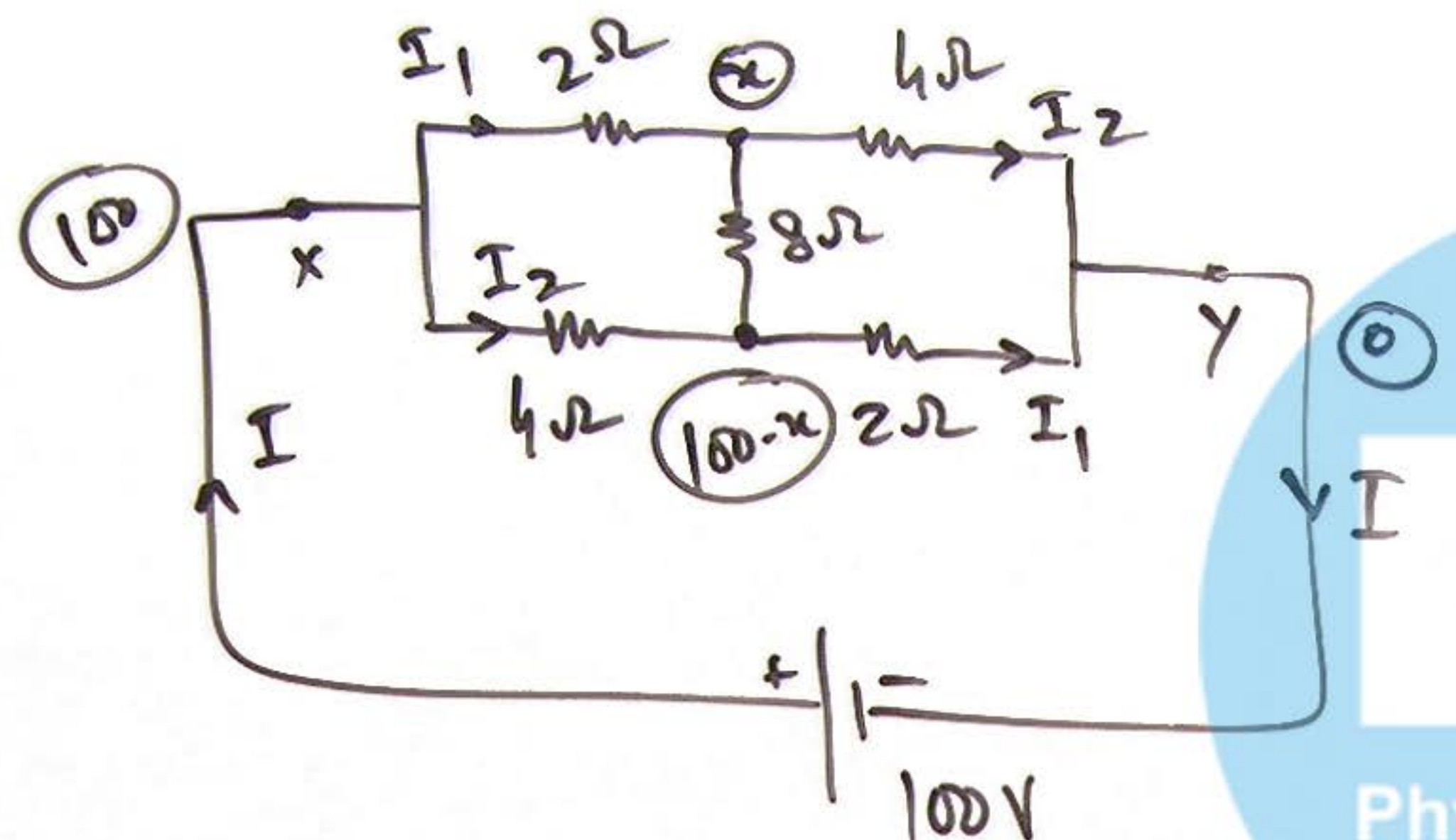


⑭ Ladder Networks:



$$\Rightarrow R_e = 3 + \frac{2R_e}{2 + R_e} \Rightarrow R_e = \checkmark$$

15 Symmetry Circuits:



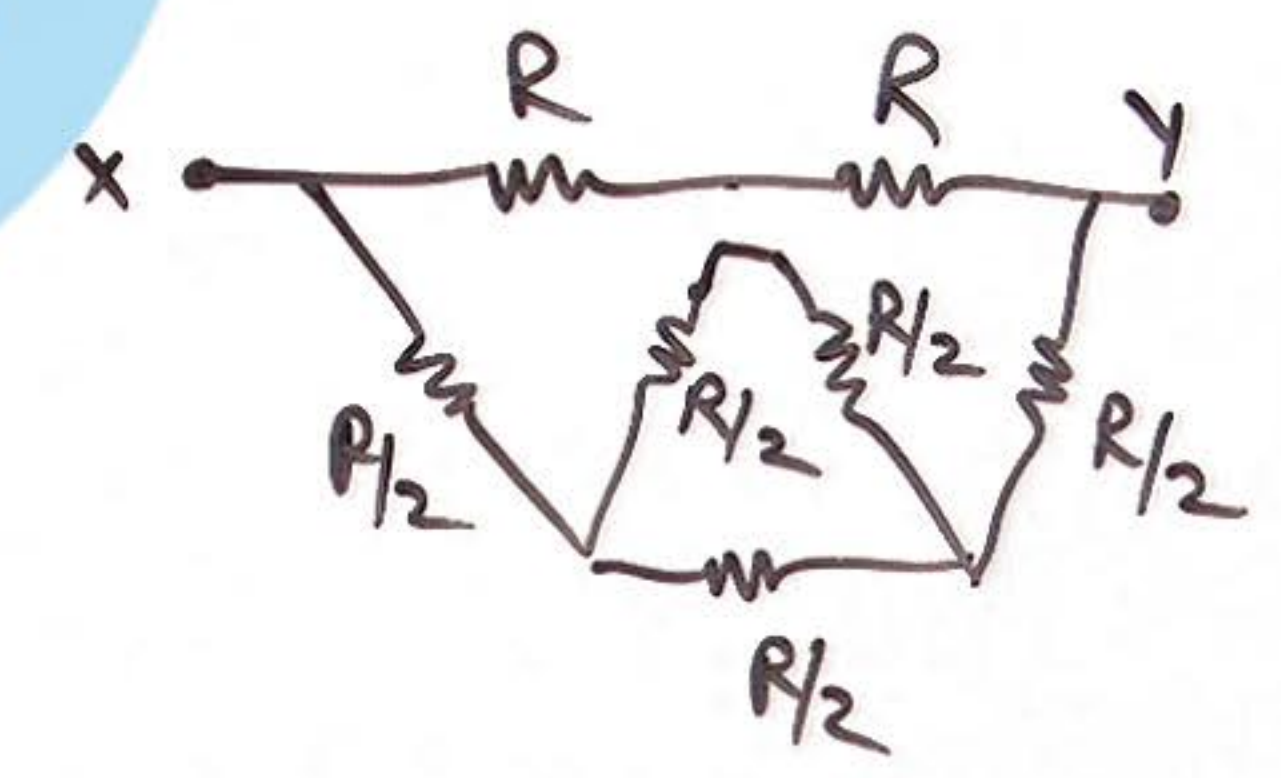
$$R_{eq} = \frac{100}{I}$$

$$\Rightarrow R_{eq} = \frac{100}{\frac{x}{4} + \frac{100-x}{2}}$$

KCL eqⁿ of $\underline{x}$

$$\frac{x}{4} + \frac{2x-100}{8} + \frac{x-100}{2} = 0$$

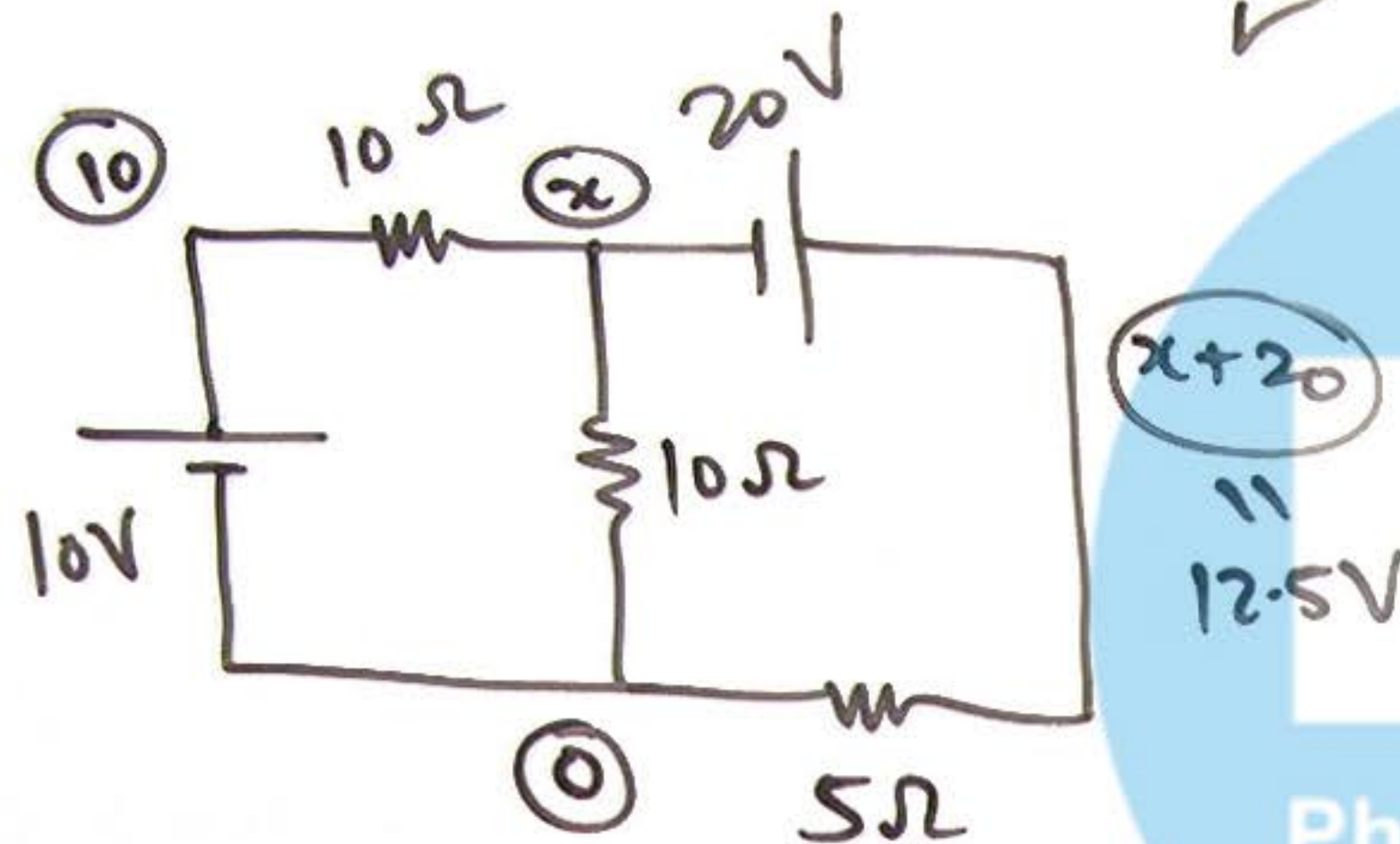
$$x = \underline{\hspace{2cm}}$$



16) Circuits with Two or more Batteries:

Q1: Find $I_{5\Omega}$?

$$I_{5\Omega} = \frac{12.5}{5} = 2.5A$$



KCL eqⁿ for x-

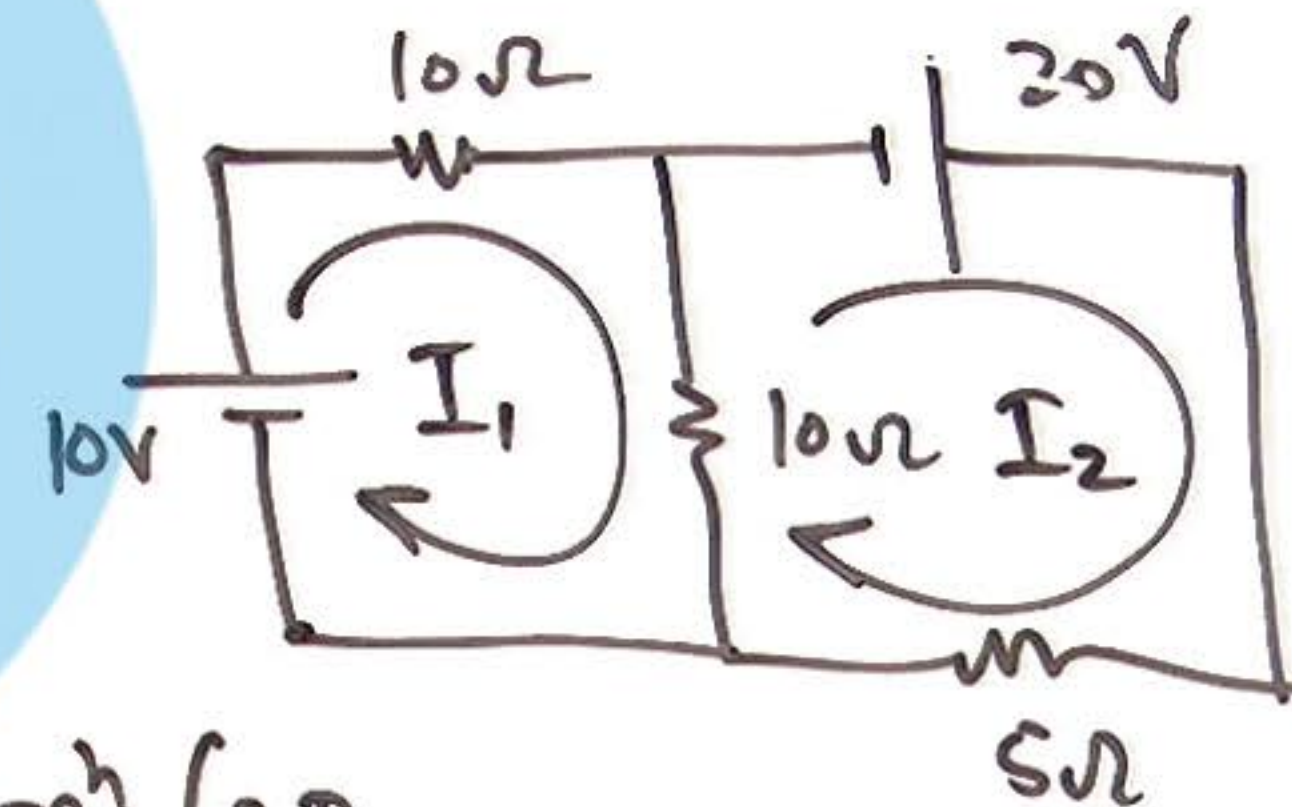
$$\frac{x-10}{10} + \frac{x}{10} + \frac{x+20}{5} = 0$$

$$x-10+x+2x+40=0 \Rightarrow x=-7.5V$$

17) Kirchoff's Voltage Law: (KVL)

KVL is applied for a loop

$$\sum_{i=1}^n I_i R_i = \sum_{j=1}^m e_j$$



KVL eqⁿ for I_1 -

$$+10 - 10I_1 - 10(I_1 - I_2) = 0 \quad \text{--- (1)}$$

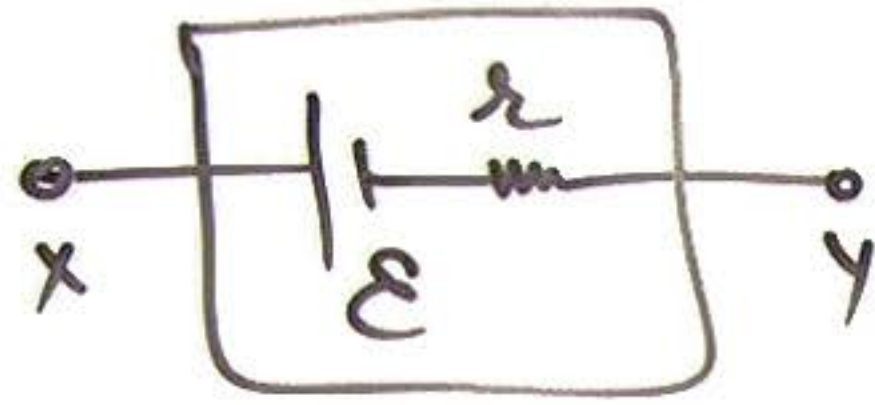
KVL eqⁿ for I_2 -

$$+20 - 5I_2 - 10(I_2 - I_1) = 0 \quad \text{--- (2)}$$

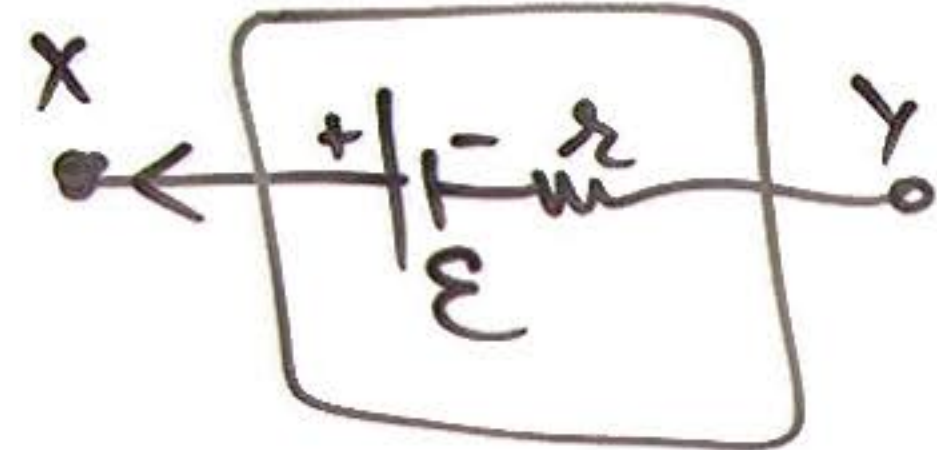
from (1) & (2)
 $I_2 = \checkmark$

18 EMF & Cells grouping:

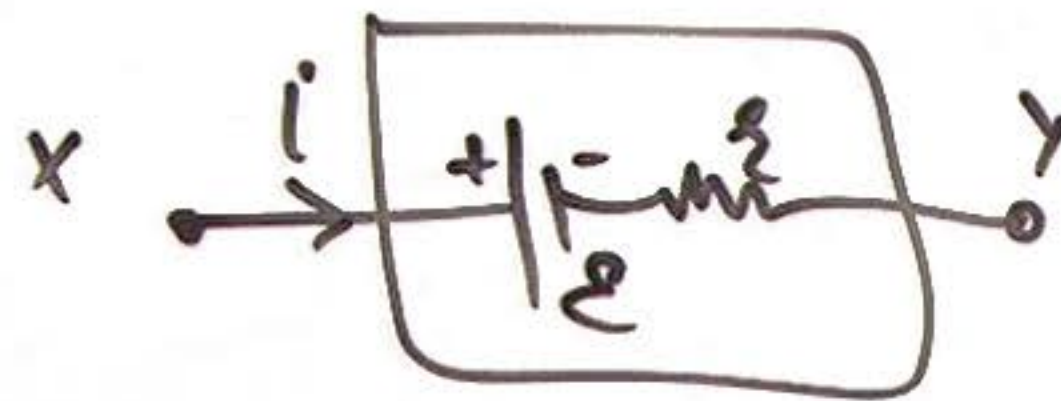
Open circuit
p.d. across a
Battery Terminal



$$i = 0 \Rightarrow V_x - V_y = \mathcal{E}$$

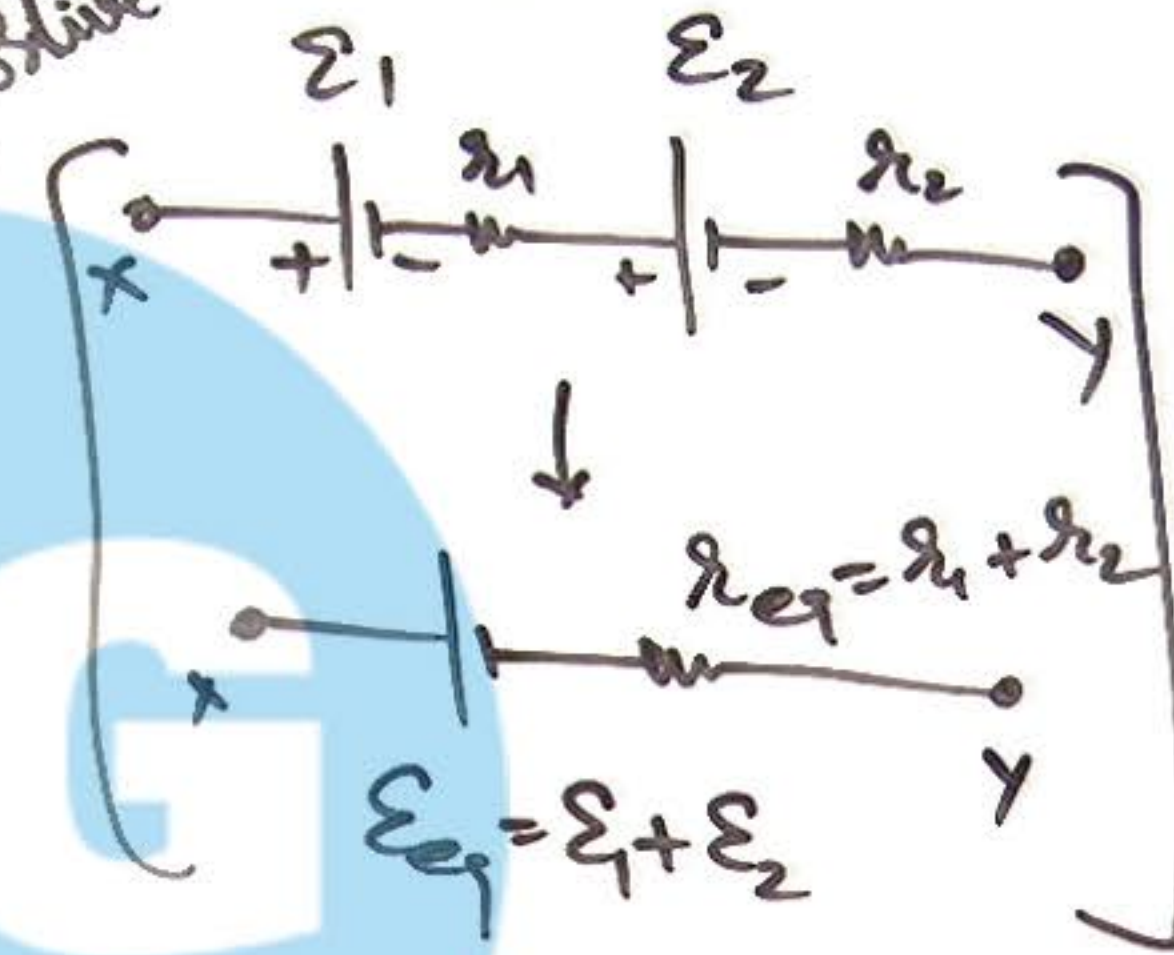


$$\Rightarrow V_x - V_y = \mathcal{E} - ir$$

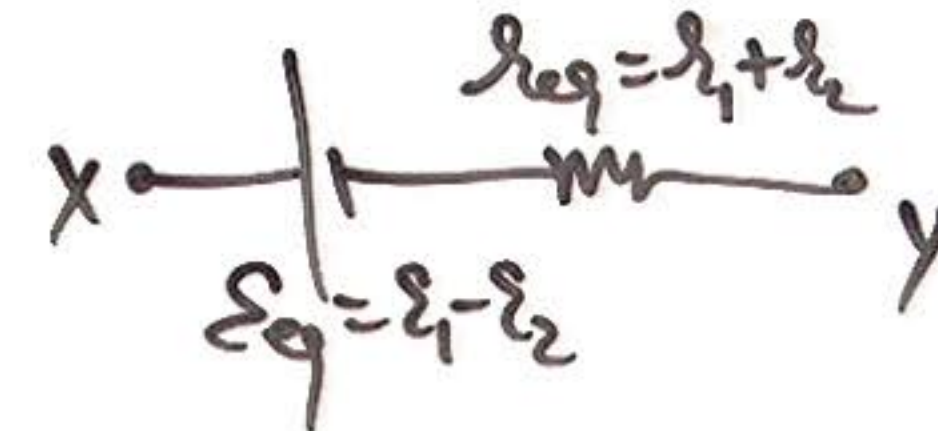
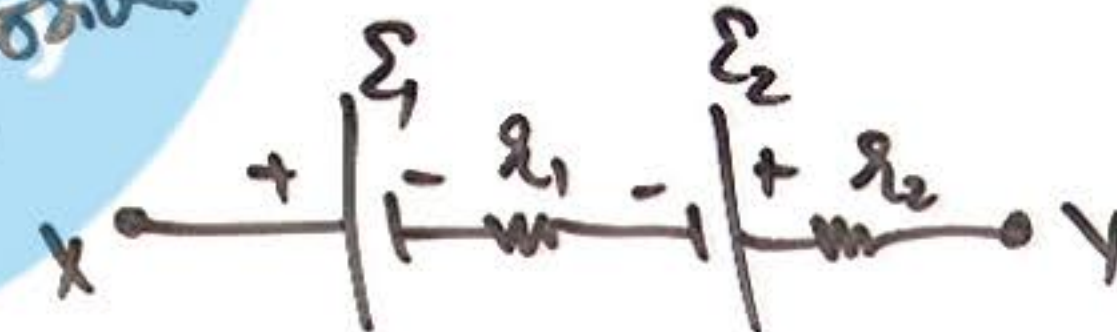


$$\Rightarrow V_x - V_y = \mathcal{E} + ir$$

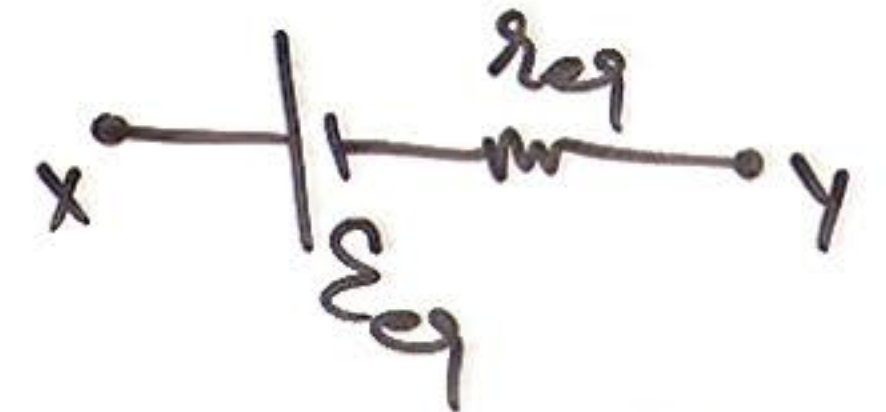
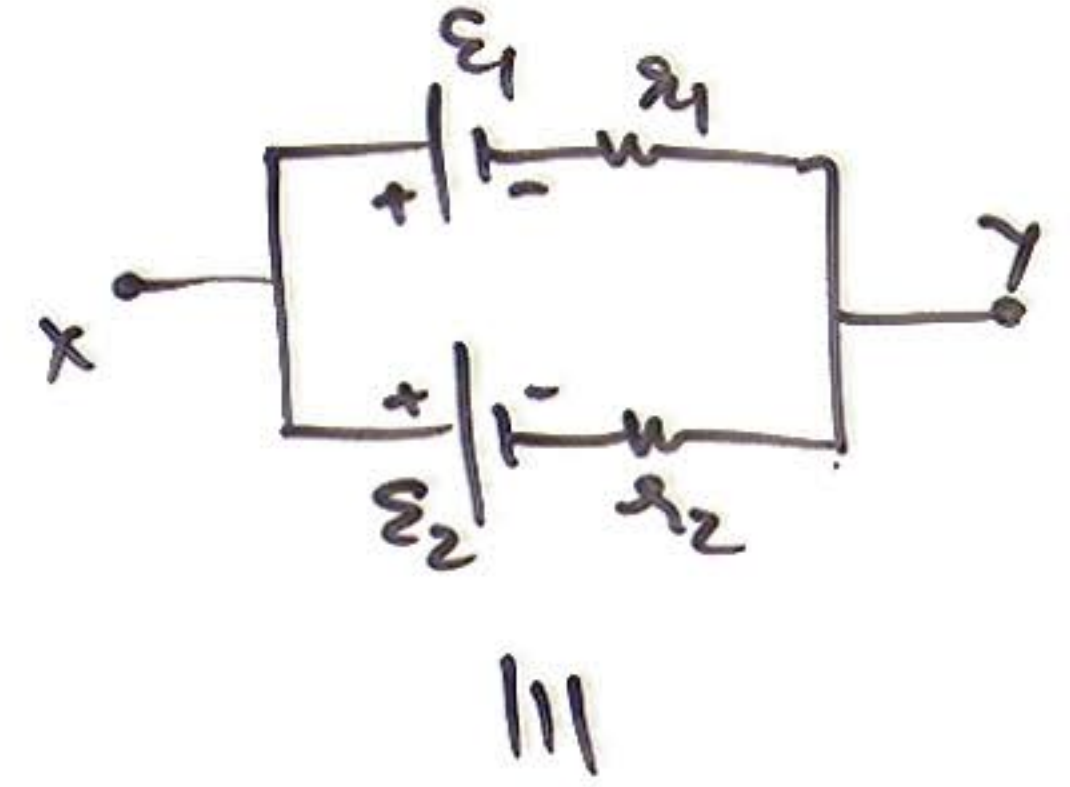
Supportive Series



Opposite



Parallel



$$\mathcal{E}_eq = \frac{\mathcal{E}_1/r_1 + \mathcal{E}_2/r_2 + \dots}{1/r_1 + 1/r_2 + \dots}$$

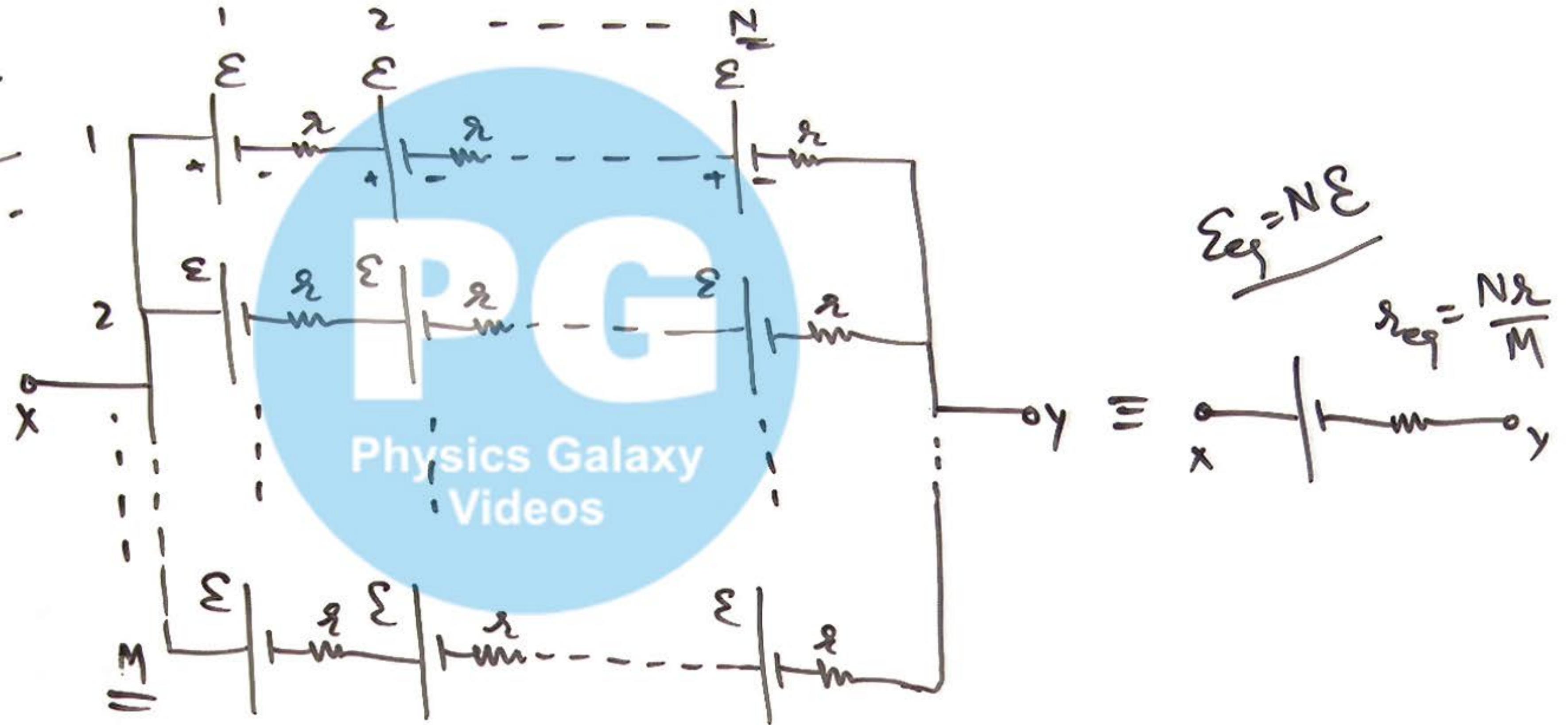
$$\frac{1}{r_eq} = \frac{1}{r_1} + \frac{1}{r_2} + \dots$$

①9 Battery Grid:

$$\Sigma_{eq} = \frac{\frac{\Sigma_1}{r_1} + \frac{\Sigma_2}{r_2} + \dots}{\frac{1}{r_1} + \frac{1}{r_2} + \dots}$$

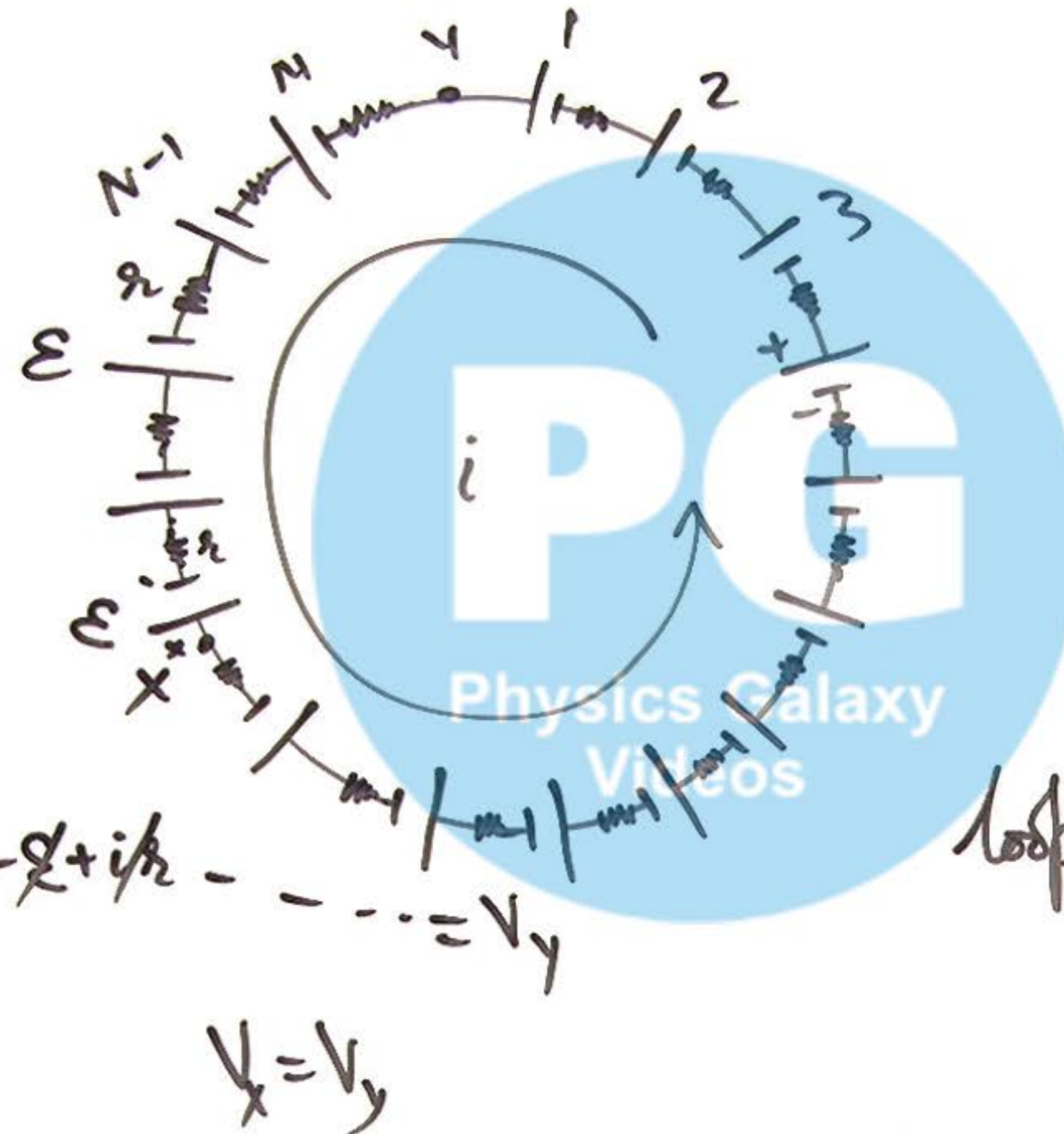
$$\underline{\underline{\Sigma_1 = \Sigma_2 = \Sigma}}$$

$$\Rightarrow \Sigma_{eq} = \Sigma$$



(20) Loop of identical Batteries:

$$\underline{V_x - V_y = 0.}$$



for loop

$$R_{eq} = Nr$$

$$\Sigma \mathcal{E} = N\mathcal{E}$$

$$\text{loop current } i = \frac{\Sigma \mathcal{E}}{R_{eq}} = \frac{\Sigma}{r}$$

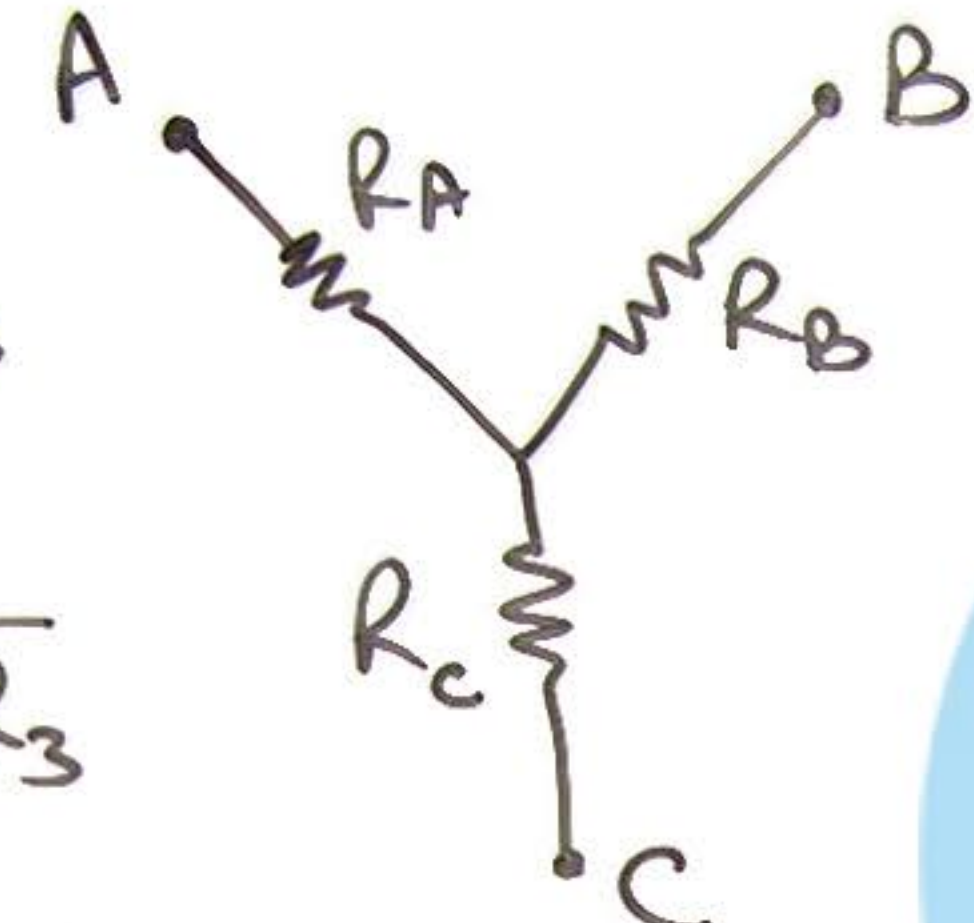
$$V_x - \mathcal{E} + ir - \mathcal{E} + ir - \dots = V_y$$

$$V_x = V_y$$

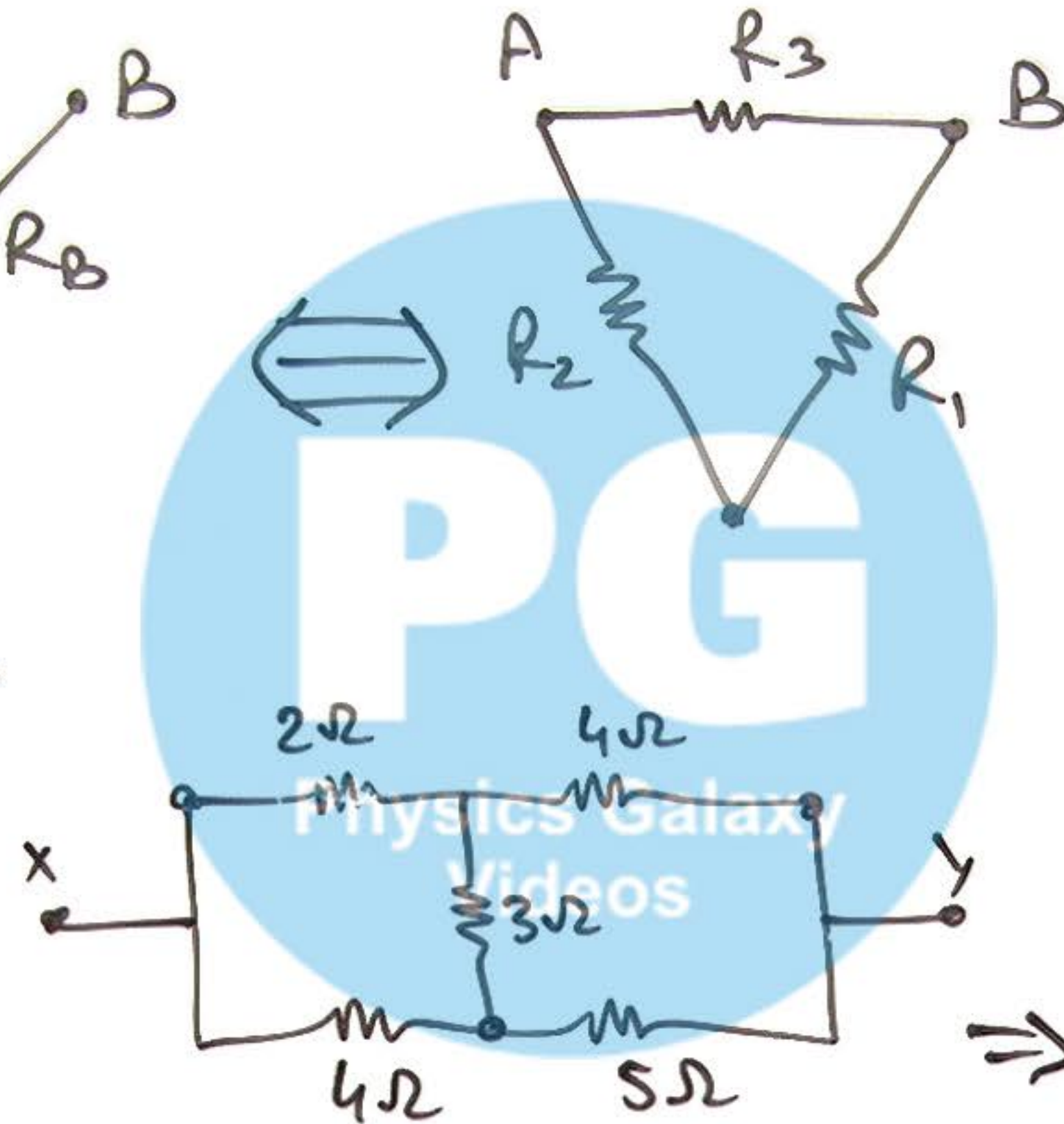
21 Δ -Y Transformation:

$$R_A = \frac{R_2 R_3}{R_1 + R_2 + R_3}$$

$$R_B = \frac{R_1 R_3}{R_1 + R_2 + R_3}$$

$$R_C = \frac{R_1 R_2}{R_1 + R_2 + R_3}$$


IL^o



$$R_1 = \frac{R_A R_B + R_B R_C + R_A R_C}{R_A}$$

$$R_2 = \frac{R_A R_B + R_B R_C + R_A R_C}{R_B}$$

$$R_3 = \frac{R_A R_B + R_B R_C + R_A R_C}{R_C}$$

